

S1A

Permutations and Combinations

Learning Objectives

In this chapter, we will

- (a) understand and use the addition and multiplication principles for counting;
- (b) differentiate between permutations and combinations;
- (c) understand that nC_r is the number of combinations of r objects from n distinct objects;
- (d) understand that nP_r is the number of permutations of r objects taken from n distinct objects; and that in the special case when $r = n$, the number of permutations of n distinct objects arranged in a straight line is $n!$;
- (e) find the number of permutations of n distinct objects arranged in a circle;
- (f) find the number of permutations of n objects which are not all distinct, e.g. if m of the objects are identical and the rest are distinct, the number of permutations is $\frac{n!}{m!}$;
- (g) interpret the relation ${}^nP_r = {}^nC_r \times r!$, and use the formula:

$${}^nC_r = \frac{n(n-1)(n-2)\cdots(n-r+1)}{r!} = \frac{n!}{r!(n-r)!};$$
- (h) use nP_r , $n!$ and nC_r to solve counting problems, including cases involving repetition and restriction.

1.1 Introduction

There are many instances in which we exercise counting to help us make decisions, for example:

- the number of possible routes from home to school using a single mode or a combination of different modes of transport
- the number of ways to select three recipes from a cookbook containing 12 recipes
- the number of ways to match six different coloured shirts to three different coloured pants
- the number of pairings between five different necklaces and 14 different ear rings

The counting problem is one of the basic problems in combinatorics: a branch of mathematics which seeks to determine the number of different arrangements (permutations) or selections (combinations) in a particular situation. In this chapter, we shall learn the concepts of permutations and combinations and apply them in solving basic counting problems.

1.2 Basic Counting Principles

In solving counting problems, there are two basic principles that are commonly used. They are the Addition Principle and the Multiplication Principle.

The Addition Principle

Suppose that E_1 and E_2 are two events that are mutually exclusive (i.e., cannot both occur at the same time). If E_1 can occur in m different ways, and E_2 can occur in n different ways, then either E_1 or E_2 can occur in $(m + n)$ ways.

🔍 Note

1. The Addition Principle can be extended to any finite mutually exclusive events.
2. The term ‘mutually exclusive’ will be dealt with in greater detail in 1.15.

Example 1

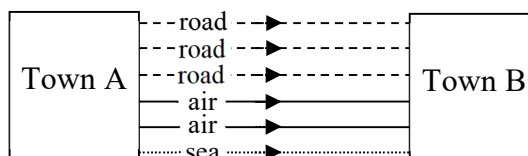
One can travel from Town A to Town B by road, air and sea. Suppose that there are three ways by road, two ways by air and one way by sea from Town A to Town B. What is the total number of ways from Town A to Town B by road, air or sea?

Solution

Number of ways by road = 3

Number of ways by air = 2

Number of ways by sea = 1

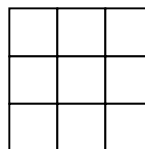


By Addition Principle, total no. of ways = $3 + 2 + 1 = 6$ □

Since the different routes are mutually exclusive.

Example 2

Find the number of squares contained in an 3×3 array where each cell is a square.



Solution

Type of square	No. of squares
1×1	9
2×2	4
3×3	1

There are only 3 types of squares in an 3×3 array:

- 1×1 square
- 2×2 square
- 3×3 square

By Addition Principle, total no. of squares = $9 + 4 + 1 = 14$

□

The Multiplication Principle

Suppose that E_1 and E_2 are 2 events that occur in sequence, i.e., one after another. If E_1 can occur in m different ways, and E_2 can occur in n different ways, then E_1 and E_2 can occur in succession in $(m \times n)$ ways.

Note

The Multiplication Principle can be extended to any finite number of events.

Example 3

In how many ways can two pupils of different gender be selected from a group of four boys and three girls?

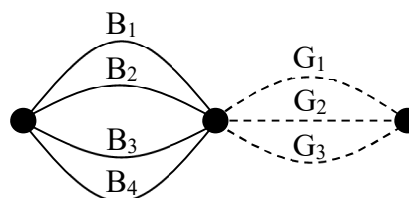
Since selection is just 2 pupils where order of selection is not a requirement, we can select a boy followed by a girl or vice versa.

Solution

Number of ways to select a boy = 4

Number of ways to select a girl = 3

By Multiplication Principle,
total no. of ways = $4 \times 3 = 12$



□

Example 4

A simple 5-character security code is made up of two different letters of the alphabet in upper case, followed by 3 digits (which may or may not be the same). An example of such a security code is AB022. Find the number of 5-character codes that can be formed.

5-character codes can be formed by considering the number of possible ways for each letter and digit in sequence.

Solution

	1 st letter	2 nd letter	1 st digit	2 nd digit	3 rd digit
No. of ways	26	25	10	10	10

By Multiplication Principle,

total no. of 5-character codes = $26 \times 25 \times 10 \times 10 \times 10 = 650000$

□

Example 5

Find the number of 3-digit numbers that can be formed from the set $\{1, 2, 3, 4, 5, 6\}$ given that the digits may be repeated.

Solution

Consider each of the digits in sequence:

	hundreds	tens	ones
No. of possible digits	6	6	6

By Multiplication Principle, total number of 3-digit numbers $= 6 \times 6 \times 6 = 6^3 = 216$.

1.3 Permutations

Suppose we want to find out the number of different ways of arranging the letters A , B and C in a row. The number of possible ways can easily be obtained by listing the letters systematically to obtain the following arrangements:

$ABC \quad ACB \quad BAC \quad BCA \quad CAB \quad CBA$

A permutation of a set of objects is an arrangement of these objects where the order of the objects is important.

1.4 # Permutations of n Distinct Objects

We have seen that listing objects systematically is one possible method of counting the number of permutations in a given problem. However, systematic listing becomes tedious and impractical when the number of objects is large.

Let us consider arranging in sequence the letters A , B and C in a row:

	1 st letter	2 nd letter	3 rd letter
No. of ways	3	2	1

Therefore, by the Multiplication Principle, total number of ways $= 3 \times 2 \times 1 = 6$.

In general, if n distinct objects are to be arranged in a row, then

	1 st object	2 nd object	3 rd object	...	$(n-1)^{\text{th}}$ object	n^{th} object
No. of ways	n	$n-1$	$n-2$	...	2	1

Number of permutations (or arrangements) of n distinct objects is given by

$$n(n-1)(n-2)\cdots(3)(2)(1) = n! .$$

Note

- The symbol ' $!$ ' represents factorial. We read ' $n!$ ' as ' n factorial'.
- By definition, $0! = 1$.

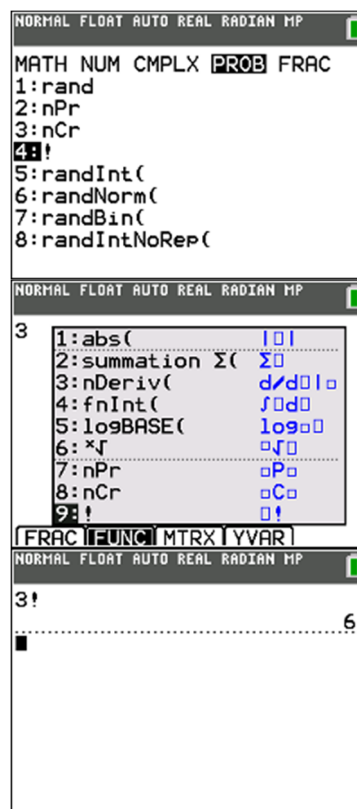
Using TI84+ to evaluate 3!:

- ① Key in 3.
- ② Press **[MATH]** > **PROB**, select **4: !**.

OR

- ② Press **[alpha]**, **[window]**, select **9: !**.

- ③ Press **[ENTER]** to evaluate.



Example 6

Five men and five women sign up for an online dating service. In how many ways can they be paired-up?

Solution

Consider pairing the five men to each of the five women in sequence.

	1 st woman	2 nd woman	3 rd woman	4 th woman	5 th woman
No. of ways to select a man	5	4	3	2	1

Total no. of ways = $5! = 120$

This is similar to arranging five distinct objects in a row.

□

1.5 # Arrangements of n Objects (Not All Distinct)

In real life, we may encounter situations where some of the objects are identical. Suppose we want to find out the number of 3-digit numbers that can be formed by the set $\{1, 1, 2\}$. We will use '1_a' and '1_b' to differentiate the two '1's' first. By listing the numbers, we have the following cases:

1_a1_b2 1_b1_a2 1_b21_a 1_a21_b 21_a1_b 21_b1_a

Since '1_a' and '1_b' are essentially the same, therefore number of 3-digit numbers

$$= \frac{3!}{2!} = 3.$$

two '1's, which are the same.

In a collection of n objects, if n_1 are of type 1, n_2 are of type 2, n_3 are of type 3, ... , n_k are of type k , and $n = n_1 + n_2 + n_3 + \dots + n_k$, then the number of arrangements of these n objects in a row is given by

$$\frac{n!}{n_1! n_2! n_3! \dots n_k!}.$$

(See Appendix for a proof for the above result.)

Example 7

In how many ways can the letters of the word 'STATISTICS' be arranged?

Solution

S	S	S	T	T	T	I	I	A	C
---	---	---	---	---	---	---	---	---	---

Total no. of ways = $\frac{10!}{3!3!2!} = 50400$

There are 10 letters consisting of 3 'S', 3 'T' and 2 'I'. □

Example 8

A bag contains five red and three blue tokens, all of which are identical, apart from colour. Four tokens are taken out of the bag at random without replacement and arranged in a row. In how many ways can this be done?

Solution

There are 4 mutually exclusive possible cases.

	Tokens	No. of ways
Case 1	3 blue, 1 red	$\frac{4!}{3!} = 4$
Case 2	2 blue, 2 red	$\frac{4!}{2!2!} = 6$
Case 3	1 blue, 3 red	$\frac{4!}{3!} = 4$
Case 4	4 red	$\frac{4!}{4!} = 1$

Hence total no. of ways = $4 + 6 + 4 + 1 = 15$ □

1.6 # Combinations

Consider the previous example of finding out the number of ways of arranging the letters A, B and C in a straight line. We know that in arranging the letters, the order of selection is important. Thus, the ways of arranging the letters are:

$$ABC \quad ACB \quad BAC \quad BCA \quad CAB \quad CBA$$

and the number of arrangements is 6.

However, if we are not arranging the letters, the order of selection is not important. Then there is only one way to select the three letters A, B and C , i.e. simply $\{A, B, C\}$.

A combination from a set of objects is a selection of a number of these objects in which the order of selecting the objects in the set is not important.

1.7 Combinations of r Objects taken from a set of n Distinct Objects

Suppose we want to find the number of combinations in selecting r objects from a set of n distinct objects.

Number of combinations (or selection) of r objects taken from n distinct objects, is given by

$${}^nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

We will illustrate the above result in greater detail in 1.9.

Note

- We use the notation nC_r or $\binom{n}{r}$, read as ‘ n choose r ’, to denote the number of combinations (or selections) of r objects taken from n distinct objects. $\binom{n}{r}$ is

NOT to be mistaken as a 2×1 column vector.

- Since $\binom{n}{r} = \frac{n!}{r!(n-r)!}$, and $\binom{n}{n-r} = \frac{n!}{(n-r)!(n-(n-r))!} = \frac{n!}{(n-r)!r!}$, we have

$$\binom{n}{r} = \binom{n}{n-r}.$$

We may interpret this result as the number of ways of selecting r objects from n distinct objects is the same as the number of ways of choosing the $(n-r)$ objects from n distinct objects.

Examples: $\binom{7}{3} = \binom{7}{4}$; $\binom{10}{2} = \binom{10}{8}$

Example 9

There are 13 female and 7 male students in a class.

- (a) A 4-member committee is to be selected. In how many ways can the committee be selected if
- there is no restriction to the selection?
 - there must be at least two female students in the committee?
- (b) In how many ways can the 20 students be divided into two groups of
- 8 and 12 students each?
 - 10 students each?

Solution

(a)(i) Required no. of ways = ${}^{20}C_4 = 4845$

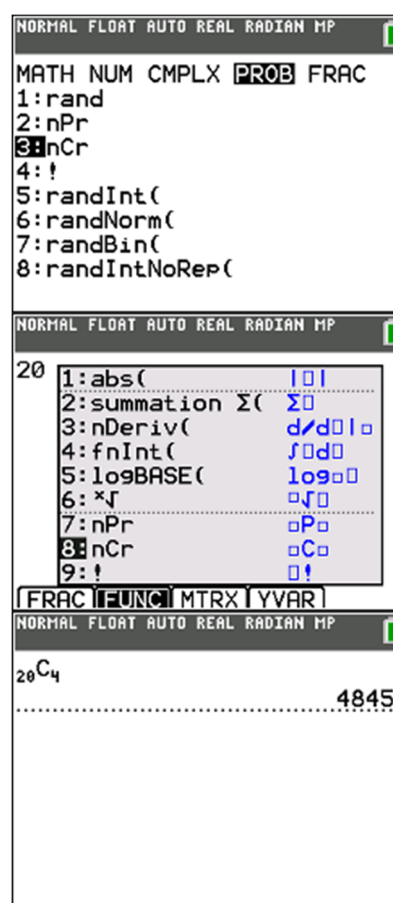
Using TI84+ to evaluate ${}^{20}C_4$:

- Key in 20.
- Press **MATH** > **PROB**, select **3:nCr**, Press **ENTER**.
- Key in 4.

OR

- Press **alpha**, **window** select **8:nCr**, Press **ENTER**.
- Key in 4.

- Press **ENTER** to evaluate.



(ii) Method 1: (consider cases)

	Committee	No. of ways
Case 1	2 females, 2 males	${}^{13}C_2 \times {}^7C_2 = 1638$
Case 2	3 females, 1 male	${}^{13}C_3 \times {}^7C_1 = 2002$
Case 3	All 4 females	${}^{13}C_4 = 715$

$$\therefore \text{Required no. of ways} = ({}^{13}C_2 \times {}^7C_2) + ({}^{13}C_3 \times {}^7C_1) + ({}^{13}C_4) = 4355$$

Method 2: (using method of ‘Complementation’)

Consider a committee with at most 1 female student:

	Committee	Number of ways
Case 1	no female, all 4 males	${}^7C_4 = 35$
Case 2	1 female, 3 males	${}^{13}C_1 \times {}^7C_3 = 455$

$$\begin{aligned} \therefore \text{Required no. of ways} &= \text{no. of ways with no restriction} \\ &\quad - \text{no. of ways with at most 1 female student} \\ &= {}^{20}C_4 - \left[{}^7C_4 + ({}^{13}C_1 \times {}^7C_3) \right] = 4845 - 490 = 4355 \end{aligned}$$

(b)(i) Required no. of ways $= {}^{20}C_8 \times {}^{12}C_{12} = 125970$

If we choose 8 out of 20 students to form a group, remaining 12 students would form another group automatically, ie. ${}^{12}C_{12} = 1$ way. Hence it suffices to find ${}^{20}C_8$. Note that since ${}^{20}C_8 = {}^{20}C_{12}$, we can also calculate ${}^{20}C_{12}$ instead.

(ii) Required no. of ways $= \frac{{}^{20}C_{10}}{2!} = 92378$

Since both groups have same group size 10, giving the answer as ${}^{20}C_{10}$ would result in double-counting and is thus incorrect. How does double-counting occur?

	Group	Group
Selection 1	$S_1, S_2, \dots, S_{10}$	$S_{11}, S_{12}, \dots, S_{20}$
Selection 2	$S_{11}, S_{12}, \dots, S_{20}$	$S_1, S_2, \dots, S_{10}$

For example, in Selection 1, students $S_1, S_2, \dots, S_{10}$ are selected to form a group, resulting in students $S_{11}, S_{12}, \dots, S_{20}$ to be in another group automatically. In Selection 2, students $S_{11}, S_{12}, \dots, S_{20}$ are now selected to form a group, resulting in students $S_1, S_2, \dots, S_{10}$ to be in another group automatically. Notice that both Selections are essentially the same with $S_1, S_2, \dots, S_{10}$ in one group and $S_{11}, S_{12}, \dots, S_{20}$ in another, but they are counted as 2 different ways! Thus there is double-counting, and we need to divide ${}^{20}C_{10}$ by 2!.

Note that if groups are labelled, eg. Group A and Group B, then ${}^{20}C_{10}$ is correct. There is no need to divide by 2! since $S_1, S_2, \dots, S_{10}$ in Group A and $S_{11}, S_{12}, \dots, S_{20}$ in Group B is different from $S_1, S_2, \dots, S_{10}$ in Group B and $S_{11}, S_{12}, \dots, S_{20}$ in Group A.

In general, if there are k groups with same group size, there is a need to divide by $k!$ to remove multiple-counting.

Eg. To form 3 groups of 4 from 12 people, no. of ways $= \frac{{}^{12}C_4 \times {}^8C_4 \times {}^4C_4}{3!} = 5775$.

1.8 # Arrangements of r Objects taken from a set of n Distinct Objects

Suppose r objects are to be taken from a set of n distinct objects and arranged in a row. Each object, once taken, will not be considered again. Thus, each arrangement is also a permutation of the chosen r objects. Arranging the r objects in sequence:

	1 st object	2 nd object	3 rd object	...	$(r-1)^{\text{th}}$ object	r^{th} object
No. of ways	n	$n-1$	$n-2$	...	$n-r+2$	$n-r+1$

$$\begin{aligned}
 \text{Hence number of permutations} &= n(n-1)(n-2)\cdots(n-r+1) \\
 &= \frac{n(n-1)(n-2)\cdots(n-r+1)(n-r)(n-r-1)\cdots 3\cdot 2\cdot 1}{(n-r)(n-r-1)\cdots 3\cdot 2\cdot 1} \\
 &= \frac{n!}{(n-r)!}
 \end{aligned}$$

Number of permutations of r objects taken from n distinct objects is given by

$${}^nP_r = n(n-1)(n-2)\cdots(n-r+1) = \frac{n!}{(n-r)!}.$$

Note

- We use the notation nP_r to denote the number of permutations of r objects taken from n distinct objects.
- As discussed previously in 1.4, the number of permutations of n distinct objects is ${}^nP_n = n(n-1)(n-2)\cdots 3\cdot 2\cdot 1 = n!$.

Example 10

A man has 10 different pieces of jewellery. He is to select 3 pieces and give one piece each to his three daughters as dowry. In how many ways can he do this?

Solution

This is similar to a permutation of three objects taken from 10 distinct objects and so we may also use the formula directly as follows:

$$\text{No. of ways} = {}^{10}P_3 = \frac{10!}{7!} = 10 \times 9 \times 8 = 720$$

OR

	1 st daughter	2 nd daughter	3 rd daughter
No. of ways	10	9	8

Consider each daughter in sequence.

By Multiplication Principle,
total no. of ways $= 10 \times 9 \times 8 = 720$

□

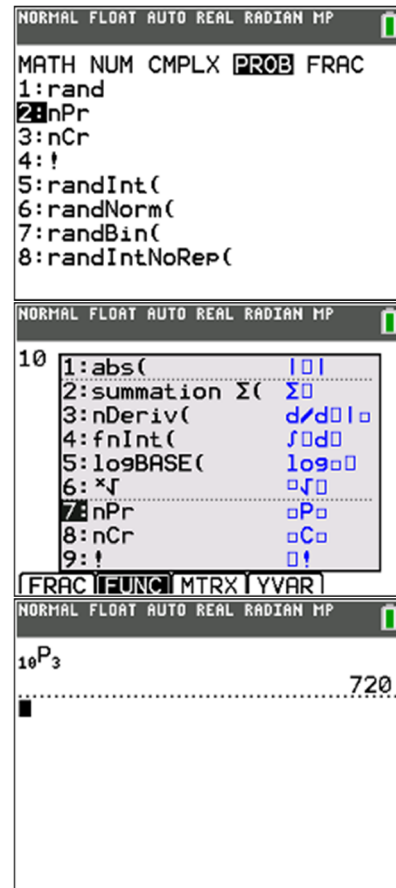
Using TI84+ to evaluate ${}^{10}P_3$:

- ① Key in 10.
- ② Press **MATH** > PROB, select **2:nPr**, Press **ENTER**.

OR

- ② Press **MATH** > PROB, select **2:nPr**, Press **ENTER**.

- ③ Key in 3.
- ④ Press **ENTER** to evaluate.



1.9 # Relationship between nC_r and nP_r

What is the relationship between a combination (or selection) and a permutation (or arrangement)? Let us revisit Example 9(a)(i). If the 4-member committee comprises a chairperson, a secretary, a treasurer and a committee member, how many ways can this committee be selected?

If we consider 20 people, taken four at a time, without replacement, then the number of ways the committee can be selected = ${}^{20}P_4 = 116280$.

We can also approach the problem by first selecting four students from the class, then assigning the position of the chairperson, secretary, treasurer and member to the four selected students.

Number of ways to select four students out of 20 = ${}^{20}C_4 = 4845$.

From the four selected students, the number of ways to elect the chairperson, secretary, treasurer and member = $4!$.

Therefore total number of ways = ${}^{20}C_4 \times 4! = 116280$.

Hence, we can see that ${}^{20}P_4 = {}^{20}C_4 \times 4! = 116280$.

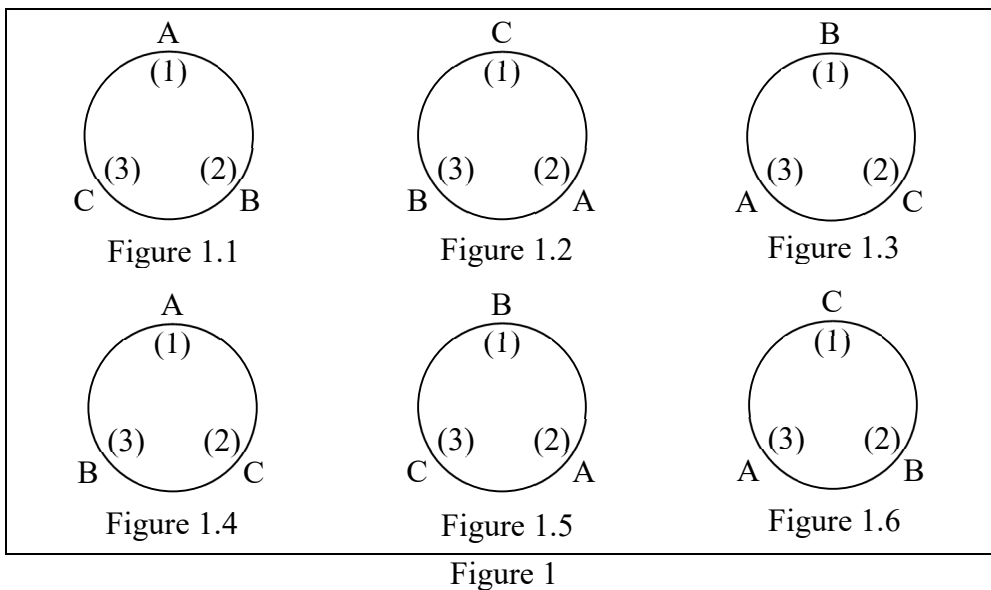
In general, we have

$${}^nP_r = {}^nC_r \times r! \quad \text{or} \quad {}^nC_r = \frac{{}^nP_r}{r!} = \frac{n!}{r!(n-r)!} .$$

1.10 # Circular Permutations

The permutations discussed so far involved objects arranged in a row. There are permutations that require arranging objects in a circle. These permutations are called circular permutations.

Consider the problem of arranging 3 distinct objects A, B, C in 3 positions in a circle. Suppose the 3 positions are numbered (1), (2) and (3). Then the possible arrangements are shown in Figure 1:



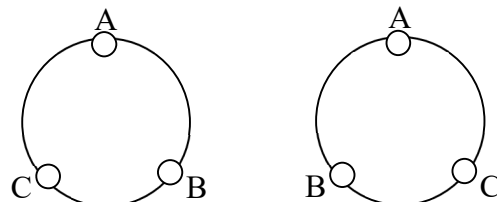
Observe that when the positions (1), (2) and (3) are disregarded, Figures 1.1, 1.2 and 1.3 are essentially the same arrangement. This is because B is always on the left of A, and C is always on the right of A. Likewise, Figures 1.4, 1.5 and 1.6 are essentially the same arrangement.

Hence, number of ways $= \frac{3!}{3} = 2$.

In general, we say that two circular permutations of the same objects are identical if any one of them can be obtained by a rotation of the other.

Alternatively, in arranging n distinct objects in a circle, we can think of it as fixing one object at a particular place in a circle to facilitate the arrangement of the rest. For instance, in arranging 3 distinct objects A, B, C in 3 positions around a circle, we fix A, and arrange B and C.

Thus number of permutations $= (3-1)! = 2! = 2$.



The number of permutations of n distinct objects in a circle is given by

$$\frac{n!}{n} = (n-1)! .$$

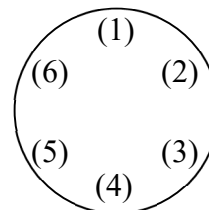
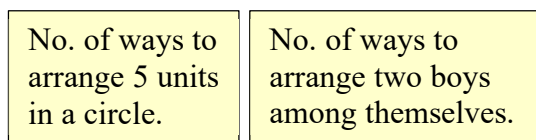
Example 11

Two boys and four girls are to be seated at a round table.

- In how many ways can this be done if there are no restrictions?
- In how many ways can this be done if the two boys must sit next to each other?

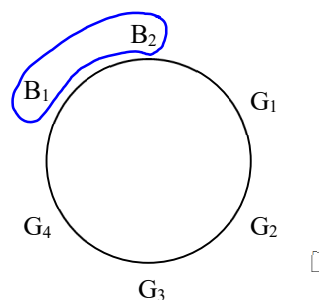
Solution

(a) No. of ways = $\frac{6!}{6} = 120$



(b) No. of ways = $\frac{5!}{5} \times 2! = 48$

Since the two boys must sit next to each other, using method of 'Grouping', group the two boys as one unit. Hence total number of units = $1 + 4 = 5$.



1.11 More Examples

Example 12

A pop artiste has five different posters of herself. In how many ways can she give away at least one of her posters?

Solution

Method 1:

Required no. of ways = $2^5 - 1 = 31$

There are 2 ways in which artiste can do to each poster: keep or give away. Hence for five posters, total no. of ways = $2 \times 2 \times 2 \times 2 \times 2 = 2^5 = 32$.

Out of the 2^5 ways, there is one in which the artiste does not give away any of her five posters.

The artiste can give away either 1, or 2, or 3, or 4, or all 5 of her posters.

Method 2:

Required no. of ways = $\binom{5}{1} + \binom{5}{2} + \binom{5}{3} + \binom{5}{4} + \binom{5}{5} = 31$

Example 13

Three boys and two girls in a class committee are to be arranged in a row in a photo-shoot.

- What is the number of ways to arrange them with no restriction?
- What is the number of ways to arrange them so that the two girls are next to each other?
- What is the number of ways to arrange them so that the two girls are not next to each other?

Solution

- No. of ways = $5! = 120$

- $\boxed{G_1 \ G_2}$
1 unit

$\boxed{B_1} \ \boxed{B_2} \ \boxed{B_3}$
← 3 units →

∴ Required no. of ways = $4! \times 2! = 48$

Since the two girls are next to each other, using method of 'Grouping', group the two girls as one unit. Hence total number of units = $1 + 3 = 4$.

No. of ways to arrange 4 units.

No. of ways to arrange two girls among themselves.

- Method 1: (using method of 'Complementation')

Required no. of ways = $120 - 48 = 72$

Note that (ii) and (iii) are complementary events.

Hence required no. of ways

= no. of ways with no restriction – no. of ways 2 girls are next to each other

= (i) – (ii)

Method 2: (using method of 'Slotting')

$\downarrow 1 \quad \downarrow 2 \quad \downarrow 3 \quad \downarrow 4$
 $\boxed{B_1} \quad \boxed{B_2} \quad \boxed{B_3}$

Since the two girls are not next to each other, use the boys as 'separators' and 'slot' the girls in between the boys.

Required no. of ways = $\binom{4}{2} \times 2! \times 3! = 72$

Note that there are 4 possible slots to place the two girls. Hence no. of ways

= 4C_2 or $\binom{4}{2}$.

No. of ways to arrange three boys among themselves.

No. of ways to arrange two girls among themselves.

Example 14

In how many ways can the letters of the word 'INSTITUTION' be arranged such that

- the first letter is 'I' and the last letter is 'N'?
- all the vowels are next to each other?
- no two vowels are next to each other?

Solution

- (i)

I	N	S	O	U	I	I	T	T	T	N
---	---	---	---	---	---	---	---	---	---	---

$$\text{Required no. of ways} = \frac{9!}{2!3!} = 30240$$

Since first 'I' and last 'N' are fixed, only need to consider the remaining 9 letters between them.

- (ii)

I	I	I	O	U
---	---	---	---	---

S

N

N

T

T

T

- 1 unit 6 units

$$\therefore \text{Required no. of ways} = \frac{7!}{2!3!} \times \frac{5!}{3!} = 8400$$

Since all the vowels are next to each other, using method of 'Grouping', group all the vowels as one unit. Hence total no. of units = $1 + 6 = 7$.
Note that there are 2 'N's and 3 'T's within the 7 units.

No. of ways to arrange the 5 vowels (with 3 'I's) among themselves.

- (iii)

↓1	↓2	↓3	↓4	↓5	↓6	↓7
S	N	N	T	T	T	

$$\text{Required no. of ways} = \binom{7}{5} \times \frac{6!}{2!3!} \times \frac{5!}{3!} = 25200$$

Note that there are 7 possible slots to place the 5 vowels. Hence no. of ways = 7C_5 or $\binom{7}{5}$.

No. of ways to arrange the 5 vowels (with 3 'I's) among themselves.

No. of ways to arrange the 6 consonants (with 2 'N's and 3 'T's) among themselves.

Since no two vowels are to be next to each other, using method of 'Slotting' with the 6 consonants as 'separators', slot the vowels in between the consonants.

⚠ Note

It is incorrect to solve (iii) by subtracting the number of ways where all the vowels are next to each other, from the number of ways to arrange the 11 letters without restriction. This incorrect method is often mistaken as the method of 'Complementation'. The number of ways obtained in this way are more than 25200, as it include cases where some but not all of the five vowels are next to each other (eg. 3 'I's together and 'O' and 'U' together but not adjacent to the 3 'I's).

Example 15

In how many ways can 4-digit integers greater than 5000 be formed using only the digits 0, 1, 3, 5, 7 and 8, if

- no digit may be repeated?
- the digits can be repeated?
- no repetition of the digits is allowed and the integers formed must be even?

Solution

(i)

Possible 4-digit integers				
	Thousands	Hundreds	Tens	Ones
No. of ways	3	5	4	3

This digit can be 5, 7 or 8.

Required no. of 4-digit integers = $3 \times 5 \times 4 \times 3 = 180$

(ii)

Possible 4-digit integers				
	Thousands	Hundreds	Tens	Ones
No. of ways	3	6	6	6

Required no. of 4-digit integers = $(3 \times 6^3) - 1 = 647$

No. of 4-digit integers that can be formed = $3 \times 6 \times 6 \times 6 = 3 \times 6^3 = 648$.

As integers greater than 5000 are required, need to exclude the integer 5000 since it is one of the 648 4-digit integers.

- (iii) Case 1: The last digit is '0'.

For an integer to be even, its last digit can only be '0' or '8', hence split into 2 cases.

Possible 4-digit integers				
	Thousands	Hundreds	Tens	Ones
No. of ways	3	4	3	1

4 digits left after '0' is fixed at the ones place, and one of '5', '7' or '8' is used at the thousands place.

'0' must be fixed here.

No. of ways = $3 \times 4 \times 3 \times 1 = 36$

Case 2: The last digit is '8'.

Possible 4-digit integers				
	Thousands	Hundreds	Tens	Ones
No. of ways	2	4	3	1

This digit can only be '5' or '7'.

'8' must be fixed here.

No. of ways = $2 \times 4 \times 3 \times 1 = 24$

Required no. of integers = $36 + 24 = 60$

□

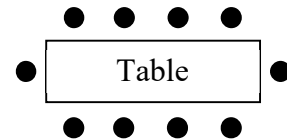
Example 16

At a wedding dinner, four single men, three single women and a married couple with a child are to be seated.

- (a) Find the number of ways of arranging these 10 guests at a round table with 10 seats if

- # (i) there are no restrictions,
- (ii) there are no restrictions with the seats numbered,
- # (iii) the child is between the mother and another woman,
- (iv) each of the three single women are not adjacent to one another, and the child is beside the mother.

- # (b) If these 10 guests are seated at a rectangular table as shown, find the number of ways of arranging them with no restrictions.



Solution

(a)(i) Required no. of ways = $\frac{10!}{10} = 362880$

Since seats are numbered, the 10 similar arrangements that arose from arrangements in a circle no longer exist as guests are seated on different numbered seats. Hence we multiply by 10, the number of seats, to remove the multiple counting.

(ii) Required no. of ways = $\frac{10!}{10} \times 10 = 3628800$

(iii) Required no. of ways

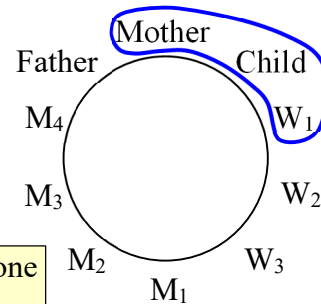
$$= \frac{8!}{8} \times \binom{3}{1} \times 2! = 30240$$

2! ways to arrange Mother and woman within unit, since they can swap places.

Choose one woman among 3 single women to be in unit with Mother and Child.

Using method of grouping, group Mother, Child and one other woman as 1 unit. $\therefore$ no. of units = $7 + 1 = 8$

No. of ways to arrange 8 units in a circle = $\frac{8!}{8}$



(iv) Required no. of ways

$$= \frac{6!}{6} \times 2! \times \binom{6}{3} \times 3! = 28800$$

3! ways to arrange 3 single women among themselves.

Using method of slotting, 6 slots are created from 6 'separators' in a circle. Choose 3 out of 6 slots to sit the 3 single women.

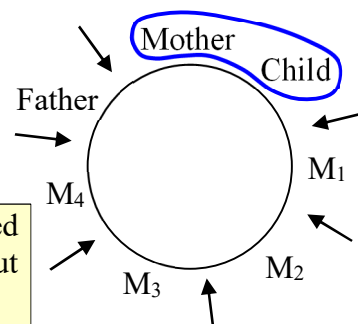
2! ways to arrange Mother and Child among themselves.

Using method of grouping, group Mother and Child as 1 unit.

Use this unit together with Father and four single men as 'separators'.

$\therefore$ no. of 'separator' units = $1 + 1 + 4 = 6$

No. of ways to arrange 6 units in a circle = $\frac{6!}{6}$



(b) Required no. of ways = $\frac{10!}{2} = 1814400$

□

The number of identical arrangements in a rectangular table is no longer the same as in a circular permutation. In Figure 1, even though the people adjacent to each guest are the same, eg. 1 is adjacent to 10 and 2, we notice that 10 is on the short edge of the table in the first arrangement but on the long edge of the table in the second arrangement. This implies that the two arrangements are not identical.

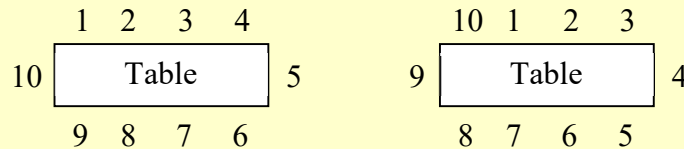


Figure 1

However, if we move all the guests seat by seat in an anti-clockwise manner until the guests on one long edge of the table are now on the other long edge of the table as shown in Figure 2, we notice that the arrangements are identical. Hence there is double counting, and instead of dividing by 10 as in arrangements in a circle, we divide by 2 to get the required number of ways.

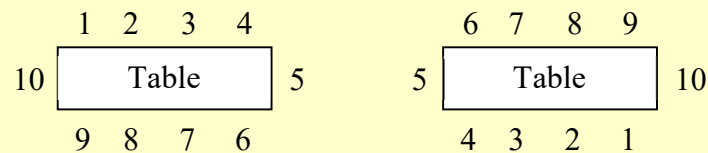


Figure 2

Summary

Basic Counting Principles	
Addition Principle Consider mutually exclusive cases	Multiplication Principle Consider consecutive operations one after another

Combination (or Selection): Order is not important	
Type of combination	Formula
n distinct objects, taken r at a time	${}^nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$

Arrangement : Order is important	
Types of arrangement	Formulae
Permutation of n distinct objects	$n!$
Permutation of r objects taken from n distinct objects, without replacement	${}^nP_r = \frac{n!}{(n-r)!} = {}^nC_r \times r!$
Arrangement of r objects taken from n distinct objects, with replacement	n^r
Arrangement of n objects, not all distinct, where $n = n_1 + n_2 + n_3 + \dots + n_k$	$\frac{n!}{n_1! n_2! n_3! \dots n_k!}$
Permutation of n distinct objects in a circle	$\frac{n!}{n} = (n-1)!$

Relationship between nP_r and nC_r :	${}^nP_r = {}^nC_r \times r!$
--	-------------------------------

Useful techniques/methods for solving counting problems: - Take note of restrictions or constraints - Consider objects in sequence - Method of ‘Complementation’ - Method of ‘Grouping’ - Method of ‘Slotting’

Appendix

In a collection of n objects, if n_1 are of type 1, n_2 are of type 2, n_3 are of type 3, ... , n_k are of type k , and $n = n_1 + n_2 + n_3 + \dots + n_k$, then the number of arrangements of these n objects in a row is given by

$$\frac{n!}{n_1! n_2! n_3! \dots n_k!}.$$

Proof:

The n_1 objects of type 1 can be placed in n_1 positions among the available n positions.

This can be done in $\binom{n}{n_1}$ ways.

We are left with $(n - n_1)$ positions.

The n_2 objects of type 2 can then be placed in n_2 positions among the remaining $(n - n_1)$ positions in $\binom{n - n_1}{n_2}$ ways.

Likewise, the n_3 objects of type 3 can be placed in n_3 positions among the remaining $(n - n_1 - n_2)$ positions in $\binom{n - n_1 - n_2}{n_3}$ ways.

Continuing in this way, we see that the n_k objects of type k can be placed in $\binom{n - n_1 - n_2 - \dots - n_{k-1}}{n_k}$ ways.

By the multiplication principle, the total number of permutations is

$$\begin{aligned} & \binom{n}{n_1} \cdot \binom{n - n_1}{n_2} \cdot \binom{n - n_1 - n_2}{n_3} \dots \binom{n - n_1 - n_2 - \dots - n_{k-1}}{n_k} \\ &= \frac{n!}{n_1! \cancel{(n - n_1)!}} \cdot \frac{\cancel{(n - n_1)!}}{n_2! \cancel{(n - n_1 - n_2)!}} \cdot \frac{\cancel{(n - n_1 - n_2)!}}{n_3! \cancel{(n - n_1 - n_2 - n_3)!}} \dots \frac{\cancel{(n - n_1 - n_2 - \dots - n_{k-1})!}}{n_k! 0!} \\ &= \frac{n!}{n_1! n_2! n_3! \dots n_k!} \end{aligned}$$

□

S1B

Probability

Learning Objectives

In this chapter, we will

- (a) understand that the probability of an event measures how likely the event will occur;
- (b) construct a table of possible outcomes to calculate probabilities; and understand that the total probability of all the possible outcomes is equal to 1;
- (c) calculate probabilities using the addition and multiplication principles;
- (d) calculate probabilities using permutations and combinations;
- (e) calculate conditional probability using $P(A|B) = \frac{P(A \cap B)}{P(B)}$;
- (f) use a Venn diagram to interpret and calculate probabilities such as $P(A')$, $P(A \cup B)$, $P(A \cap B)$ and $P(A|B)$;
- (g) use the following results: $P(A') = 1 - P(A)$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- (h) understand the meaning of mutually exclusive events, and recognize events that are, or are not, mutually exclusive through practical examples; and use the result $P(A \cup B) = P(A) + P(B)$, where A and B are mutually exclusive;
- (i) understand the meaning of independent events, and use the result $P(A \cap B) = P(A)P(B)$, where A and B are independent;
- (j) construct a tree diagram and use it to interpret and calculate probabilities, including probabilities of combined events and conditional probabilities.

1.12 # Introduction

Roulette is a casino game named after a French diminutive for little wheel (Figure 1). In the game of Roulette, players may choose to bet on either a number (0 to 36) or a range of numbers, whether the number is odd or even, or whether the number is Red or Black. The croupier then sends a small white ball spinning around the wheel which will eventually land on one of the numbers. This determines the winning number or colour. There are 18 Red numbers and 18 Black numbers. If you bet on Red, is it a fair game, i.e., is the chance (or probability) of winning equal to half? No. The casino ensures that it has an advantage by including the number '0',

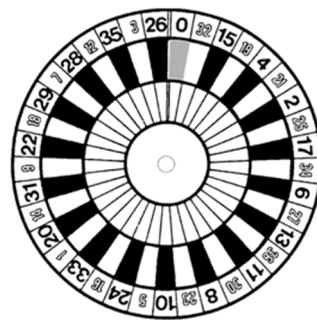


Figure 1

which is neither Red nor Black (Figure 1). The probability that you win is $\frac{18}{37}$ and the probability that the casino wins is $\frac{19}{37}$. This difference in probability may be small, but it is enough to ensure that, in the long run, the casino wins. Roulette wheels in the United States have both '0' and '00', giving the casinos an even bigger advantage. In this chapter, we will learn how to calculate probabilities using various methods.

1.13 Definitions

Definition	Example
An experiment is a situation involving chance or probability.	Rolling a die. 
An outcome is a result of an experiment.	The possible outcomes are 1, 2, 3, 4, 5 or 6.
The sample space S is the set of all possible outcomes.	Sample space $S = \{ 1, 2, 3, 4, 5, 6 \}$.
An event A is a subset of the sample space.	If A is the event of obtaining an even number, then $A = \{ 2, 4, 6 \} \subseteq S$.

If the sample space S consists of a finite number of equally likely outcomes, the probability of an event A , denoted by $P(A)$, is given by $P(A) = \frac{n(A)}{n(S)}$, where $n(A)$ denotes the number of outcomes in the event A .

🔍 Note

- $0 \leq P(A) \leq 1$ for any event A .
- If A_i , where $i = 1, 2, \dots, n$, are all possible outcomes for a sample space S , then

$$\sum_{i=1}^n P(A_i) = 1.$$
- The complement of A is denoted by A' or $\bar{A}$, i.e., A' or $\bar{A}$ is the event that ' A does not occur'.
- $P(A) + P(A') = 1$, i.e., $P(A') = 1 - P(A)$.

Example 17

In the casino game of Roulette, the table and the wheel are laid out as follows:

		0		
1 – 18	First Dozen	1	2	3
		4	5	6
Even		7	8	9
	Second Dozen	10	11	12
Red		13	14	15
		16	17	18
Black	Third Dozen	19	20	21
		22	23	24
Odd		25	26	27
	19 – 36	28	29	30
		31	32	33
		34	35	36



Figure 2

A player makes a bet by placing chips on the Roulette table (Figure 2) and the winning number (from 0 to 36) is determined by the spinning of a ball in the wheel. Of the numbers from 1 to 36, 18 of them are red and 18 are black. The number '0' is green. Find the probability that a player wins if he places a single bet on

- the pocket marked '7',
- the pocket marked 'Red',
- the pocket marked 'Second Dozen'.

Solution

(i) Required probability = $\frac{1}{37}$

All 37 numbers are equally likely.

(ii) Required probability = $\frac{18}{37}$

There are 18 red numbers.

(iii) Required probability = $\frac{12}{37}$

There are 12 numbers from 13 to 24.

□

Example 18

Two ordinary fair dice are thrown. Find the probability that

- both dice show '6',
- both dice show the same number,
- the sum of the numbers on the two dice is 8,
- the sum of the numbers on the two dice exceeds 8,
- the numbers on the two dice differ by more than 2.

Solution

(i) $P(\{(6,6)\}) = \frac{1}{36}$

(6, 6) is the only possible outcome out of 36 equally likely outcomes.

(ii) $P(\{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)\}) = \frac{6}{36} = \frac{1}{6}$

There are 6 possible outcomes where both dice show the same number, out of 36 equally likely outcomes.

(iii) $P(\{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\}) = \frac{5}{36}$

There are 5 possible outcomes where sum of two dice is 8, out of 36 equally likely outcomes.

In (i)-(iii), the **Method of Listing** is used where all possible outcomes are listed one by one.

(iv)

Sum		Number on the first die					
		1	2	3	4	5	6
Number on the second die	1	2	3	4	5	6	7
	2	3	4	5	6	7	8
	3	4	5	6	7	8	9
	4	5	6	7	8	9	10
	5	6	7	8	9	10	11
	6	7	8	9	10	11	12

$P(\text{sum of two dice exceeds 8}) = \frac{10}{36} = \frac{5}{18}$

There are 10 possible outcomes where sum of two dice exceeds 8, out of 36 equally likely outcomes.

Alternatively, using Method of Listing,
 $P(\text{sum of two dice is 9, 10, 11 or 12})$

$= P(\{(3, 6), (4, 5), (4, 6), (5, 4), (5, 5), (5, 6), (6, 3), (6, 4), (6, 5), (6, 6)\}) = \frac{10}{36} = \frac{5}{18}$

(v)

Difference		Number on the first die					
		1	2	3	4	5	6
Number on the second die	1	0	1	2	3	4	5
	2	1	0	1	2	3	4
	3	2	1	0	1	2	3
	4	3	2	1	0	1	2
	5	4	3	2	1	0	1
	6	5	4	3	2	1	0

$P(\text{numbers on two dice differ by more than 2}) = \frac{12}{36} = \frac{1}{3}$

Tabulate difference between numbers on two dice and count number of outcomes with difference more than 2.

In (iv) and (v), the **Method of Constructing a Table** is used where possible outcomes are identified from a table with all outcomes (sample space). As can be seen from (iv), the Method of Listing can become tedious as the number of possible outcomes increases. Constructing a table in such situations is more appropriate.

Concepts of probability have been around for thousands of years, but the mathematical methods of probability arose only in the mid-17th century when a simple question by a nobleman, Chevalier de Méré, to the famous mathematician Blaise Pascal (1623-1662) sparked the birth of probability theory, as we know today.

Example 19

Chevalier de Méré gambled frequently to increase his wealth. He bet on a roll of a die that at least one 6 would appear during a total of four rolls. From past experience, he knew that he was more successful than not with this game of chance. Tired of his approach, he decided to change the game. He bet that he would get a total of 12, or a double 6, on twenty-four rolls of two dice. Soon he realized that his old approach to the game resulted in more money. He asked his friend Blaise Pascal why his new approach was not as profitable.

Blaise Pascal worked out the exact probability as follows (presented here in modern notation):

Method of ‘Complementation’

Instead of calculating the probability directly which involves several cases, it is easier to **consider the complement of the required event**, which is the event of not getting any ‘6’ in four rolls of a die.

$$\begin{aligned}
 &P(\text{winning the first game}) \\
 &= P(\text{at least a ‘6’ in four rolls of a die}) \\
 &= 1 - P(\text{no ‘6’ in four rolls of a die}) \\
 &= 1 - \left(\frac{5}{6}\right)^4 \quad (\text{there are 5 out of 6 ways of no ‘6’ in 1 roll of a die}) \\
 &= 0.518
 \end{aligned}$$

Instead of finding the probability of getting a total of 12, or a double 6, in 24 rolls of two dice, Pascal found the probability of de Méré losing, or not rolling a double 6. Since there are 36 possible rolls of two dice and only one way to roll a double 6, then the probability of not rolling a double six is $\frac{35}{36}$. He rolled 24 times. The probability that

$$\text{he loses the game is: } \left(\frac{35}{36}\right)^{24} = 0.509$$

To compare to the first game, one must calculate the probability that he wins in this game.

$$P(\text{winning second game}) = 1 - \left(\frac{35}{36}\right)^{24} = 0.491$$

In subsequent centuries, many mathematicians such as Abraham de Moivre, Pierre-Simon Laplace, Daniel Bernoulli and Carl Friedrich Gauss have contributed to the theory of probability. Probability now has wide applications in diverse fields like finance, gambling, risk assessment, quantum mechanics and game theory.

1.14 # Probability Using Venn Diagrams

A **Venn diagram** is a diagram that shows all possible logical relations between a finite collection of different sets. Venn diagrams were conceived around 1880 by an English logician and philosopher John Venn (1834-1923) and are used in the fields of set theory, probability, logic, statistics, and computer science.

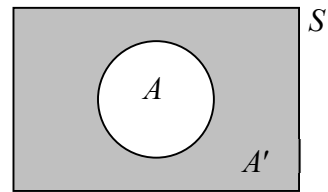
Let S be the sample space of an experiment. Let A and B be events where $A, B \subseteq S$.

The **union of A and B** , denoted by $A \cup B$, is the event that A **or** B occurs.

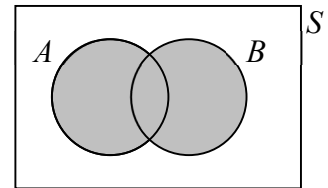
The **intersection of A and B** , denoted by $A \cap B$, is the event that both A **and** B occur.

The following illustrate some simple but important relations.

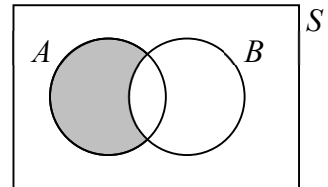
(I) $P(A') = 1 - P(A)$



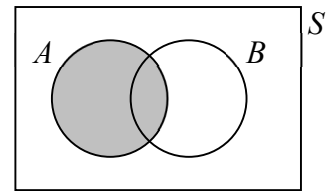
(II) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$



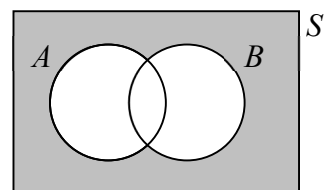
(III) $P(A \cap B') = P(A) - P(A \cap B)$
 $= P(A \cup B) - P(B)$



(IV) $P(A) = P(A \cap B) + P(A \cap B')$



(V) $P(A' \cap B') = 1 - P(A \cup B)$



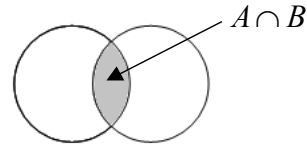
Example 20

The events A and B are such that $P(A) = 0.3$, $P(B) = 0.4$ and $P(A \cup B) = 0.5$. Find

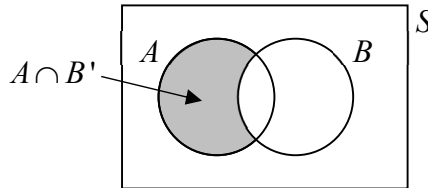
- (i) $P(A \cap B)$, (ii) $P(A \cap B')$, (iii) $P(A' \cup B)$.

Solution

$$\begin{aligned} \text{(i)} \quad P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ \therefore P(A \cap B) &= P(A) + P(B) - P(A \cup B) \\ &= 0.3 + 0.4 - 0.5 \\ &= 0.2 \end{aligned}$$

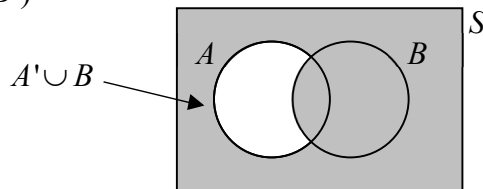


$$\begin{aligned} \text{(ii)} \quad P(A \cap B') &= P(A \cup B) - P(B) \\ &= 0.5 - 0.4 \\ &= 0.1 \end{aligned}$$



$$\begin{aligned} \text{(iii)} \quad \text{Using (ii), } P(A' \cup B) &= 1 - P(A \cap B') \\ &= 1 - 0.1 \\ &= 0.9 \end{aligned}$$

$$\begin{aligned} \text{Alternatively,} \\ P(A' \cup B) &= P(A') + P(A \cap B) \\ &= 0.7 + 0.2 \\ &= 0.9 \end{aligned}$$



□

1.15 # Mutually Exclusive Events

Two events A and B are **mutually exclusive** ($A \cap B = \emptyset$) if they cannot occur at the same time i.e., $P(A \cap B) = 0$.

For example, when a coin is tossed, the events ‘the coin shows a head’ and ‘the coin shows a tail’ are mutually exclusive.

If A and B are mutually exclusive, then they are disjoint sets as shown in the Venn diagram in Figure 3.

If A and B are mutually exclusive events, then it follows from Relation (II) in § 1.14 that $P(A \cup B) = P(A) + P(B)$, since $P(A \cap B) = 0$.

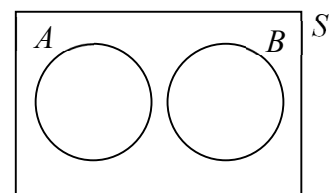


Figure 3

1.16 # Conditional Probability

The probability of an event A occurring, given that an event B has occurred, is written as $P(A|B)$. It is fair to assume here that since B has occurred, $P(B) \neq 0$.

In the Venn diagram (Figure 4), the sample space is reduced to B only, since B has occurred. That is, $P(A|B)$ is the probability of A occurring by considering B as the sample space. B is often called the reduced sample space.

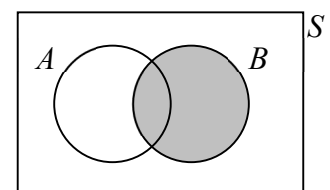


Figure 4

Thus $P(A|B)$ is the probability of the event A occurring within the sample space B , i.e., the ratio of $P(A \cap B)$ to $P(B)$.

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \text{ where } P(B) \neq 0.$$

Note

1. $P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$
2. $P(A'|B) = 1 - P(A|B)$

Example 21

Two ordinary dice are thrown. Find the probability that

- (i) the sum of the numbers on the two dice is 8, given that the numbers on the two dice are equal,
- (ii) the numbers on the two dice are equal, given that the sum of the numbers on the two dice is 8.

Solution

Let A be event that sum of two dice is 8.

Let B be event that numbers on two dice are equal.

$$(i) \quad P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(\{(4, 4)\})}{P(\{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\})} = \frac{\frac{1}{36}}{\frac{6}{36}} = \frac{1}{6}$$

OR

$$P(A|B) = \frac{n(A \cap B)}{n(B)} = \frac{n(\{(4, 4)\})}{n(\{(1, 1), (2, 2), \dots, (6, 6)\})} = \frac{1}{6}$$

$$(ii) \quad P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(\{(4, 4)\})}{P(\{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\})} = \frac{\frac{1}{36}}{\frac{5}{36}} = \frac{1}{5}$$

OR

$$P(B|A) = \frac{n(A \cap B)}{n(A)} = \frac{n(\{(4, 4)\})}{n(\{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\})} = \frac{1}{5} \quad \square$$

Note

1. $P(A|B)$ can only be evaluated using $\frac{n(A \cap B)}{n(B)}$ if **all the outcomes are equally likely**.
2. In general, for any two events A and B , $P(A|B) \neq P(B|A)$.

Question: When is $P(A|B) = P(B|A)$?

1.17 Independent Events

Two events A and B are **independent** if the occurrence of B does not affect the probability of the occurrence of A i.e., $P(A|B) = P(A)$.

In this case, we have a very special result as follows:

$$P(A) = P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

$\therefore A$ and B are independent if and only if $P(A)P(B) = P(A \cap B)$.

To show that 2 events A and B are independent, we can show either

$$P(A|B) = P(A) \text{ or}$$

$$P(B|A) = P(B) \text{ or}$$

$$P(A)P(B) = P(A \cap B).$$

Note

Two events A and B are independent $\Leftrightarrow \begin{cases} A \text{ and } B' \text{ are independent.} \\ A' \text{ and } B \text{ are independent.} \\ A' \text{ and } B' \text{ are independent.} \end{cases}$

Example 22

Draw a card at random from an ordinary deck of 52 cards.

Let A be the event of drawing an ace.

Let B be the event of drawing a spade.

Let C be the event of drawing a king.

Determine if A and B are

(i) mutually exclusive, (ii) independent.

Determine if A and C are

(iii) mutually exclusive, (iv) independent.

Solution

(i) $P(A \cap B) = \frac{1}{52} \neq 0$

Hence A and B are not mutually exclusive.

(ii) $P(A) = \frac{4}{52} = \frac{1}{13}$

Since there is only one ace out of 13 spade cards, $P(A|B) = \frac{1}{13}$.

Since $P(A|B) = P(A)$, $\therefore A$ and B are independent events.

(iii) $P(A \cap C) = 0$

Hence A and C are mutually exclusive.

(iv) $P(A) = \frac{1}{13}$ and $P(A|C) = 0$

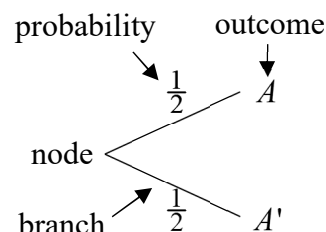
Since $P(A|C) \neq P(A)$, $\therefore A$ and C are not independent events. □

Note

1. For events A and B where $P(A) \neq 0$ and $P(B) \neq 0$, if they are mutually exclusive, we can conclude that they are not independent. Likewise if they are independent, then they cannot be mutually exclusive.
2. However, given that events A and B are not mutually exclusive, we cannot draw any conclusions about their independence. Similarly, if they are not independent, we cannot make any conclusions about their mutual exclusivity.

1.18 Probability Tree (Tree Diagram)

A probability tree or tree diagram is a useful approach when considering the probabilistic outcomes of a series of two or more events or experiments. The main parts of a probability tree are nodes (representing events) and branches (representing the outcomes of events with probabilities of occurrence of these events).



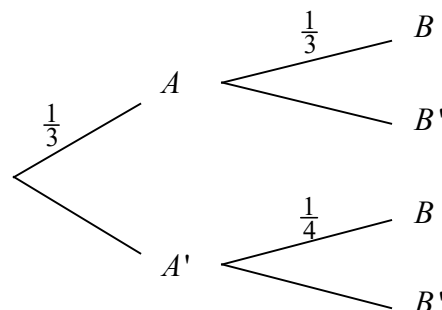
Note

At each node, there can be any number of branches. However, the probabilities on all the branches emerging from each node must add up to one.

Example 23

In an experiment involving the toss of biased coins, the probability of obtaining a 'head' on the first and second coins are $\frac{1}{3}$ and $\frac{1}{4}$ respectively.

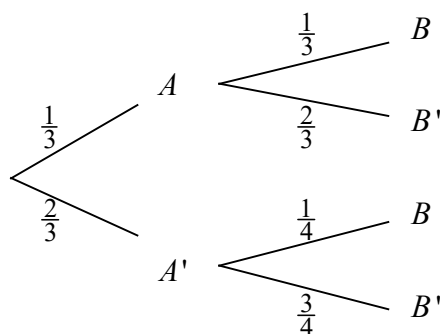
The first coin is tossed and if it shows a 'head', the first coin is tossed again. If the first toss is not a 'head', the second coin is tossed. The following is a partially completed probability tree, where A and B denote the events that a 'head' is obtained on the first and second tosses respectively.



Complete the probability tree and find

- (i) $P(B' | A)$,
- (ii) $P(B)$.

Solution



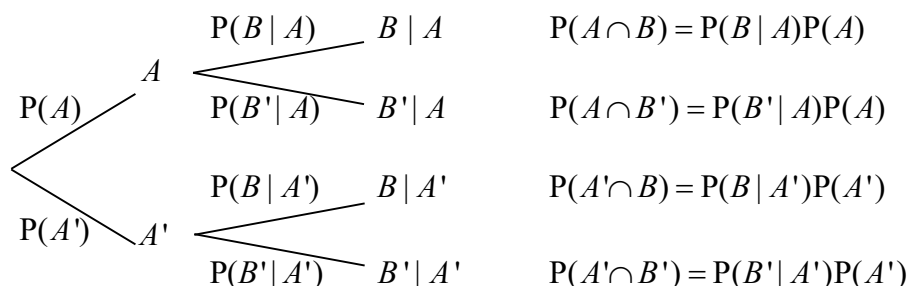
- (i) From probability tree, $P(B' | A) = \frac{2}{3}$

- (ii) $P(B) = P(B | A)P(A) + P(B | A')P(A') = \frac{1}{3}\left(\frac{1}{3}\right) + \frac{1}{4}\left(\frac{2}{3}\right) = \frac{5}{18}$

□

Note

The probabilities on branches of nodes from the second event onwards are all conditional probabilities. The events at the end of the branches are conditional events as illustrated below:



In order to simplify notations in a probability tree, events like $B | A$ or $B | A'$ are simply labelled as B since it is clear from the diagram whether A or A' has occurred.

Example 24

A bag contains two red balls and three green balls. Three balls are drawn from the bag, one after another without replacement. Find the probability that

- exactly one of the balls drawn is red,
- exactly one red ball is drawn, given that the first ball drawn is green,
- the first ball drawn is green, given that exactly one red ball is drawn.

Are the events 'exactly one of the balls drawn is red' and 'first ball drawn is green' independent?

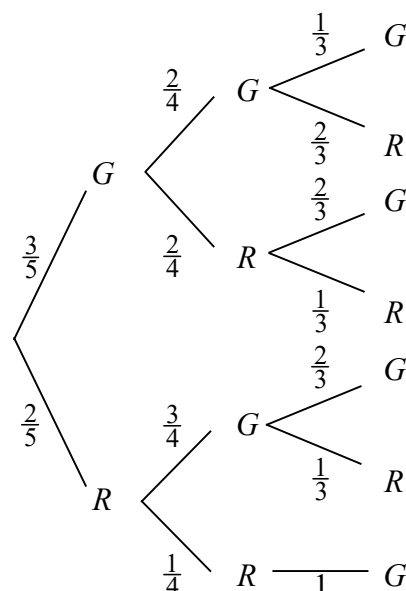
Solution

Let R be 'a red ball drawn'

Let G be 'a green ball drawn'.

$$\begin{aligned}
 \text{(i) } P(\text{exactly 1 red}) &= P(R_1 \cap G_2 \cap G_3) + P(G_1 \cap R_2 \cap G_3) \\
 &\quad + P(G_1 \cap G_2 \cap R_3) \\
 &= \left(\frac{2}{5} \times \frac{3}{4} \times \frac{2}{3}\right) + \left(\frac{3}{5} \times \frac{2}{4} \times \frac{2}{3}\right) + \left(\frac{3}{5} \times \frac{2}{4} \times \frac{2}{3}\right) \\
 &= \frac{3}{5}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) } P(\text{exactly 1 red} | 1^{\text{st}} \text{ ball green}) &= \left(\frac{2}{4} \times \frac{2}{3}\right) + \left(\frac{2}{4} \times \frac{2}{3}\right) \\
 &= \frac{2}{3}
 \end{aligned}$$



$$\begin{aligned}
 \text{(iii)} \quad & P(1^{\text{st}} \text{ ball green} \mid \text{exactly 1 red ball drawn}) \\
 &= \frac{P(1^{\text{st}} \text{ ball green and exactly 1 red ball drawn})}{P(\text{exactly 1 red ball drawn})} \\
 &= \frac{P(G_1 \cap R_2 \cap G_3) + P(G_1 \cap G_2 \cap R_3)}{\frac{3}{5}} \\
 &= \frac{\left(\frac{3}{5} \times \frac{2}{4} \times \frac{2}{3}\right) + \left(\frac{3}{5} \times \frac{2}{4} \times \frac{2}{3}\right)}{\frac{3}{5}} \\
 &= \frac{2}{3}
 \end{aligned}$$

Since $P(\text{exactly 1 red ball} \mid 1^{\text{st}} \text{ ball green}) \neq P(\text{exactly 1 red ball})$, the two events are not independent. $\square$

1.19 Miscellaneous Examples

In the calculation of probability, depending on the question, it is sometimes important to take into account the order of occurrence of the events. Consider parts (i) and (ii) of the following example.

Example 25

It is known that in a large population, 10% of the population has blood group AB. Three people are randomly chosen one at a time from this population. Find the probability that

- the first person has blood group AB and the second and third do not,
- exactly one person has blood group AB,
- at least one person has blood group AB.

Solution

$$\begin{aligned}
 \text{(i)} \quad & P(1^{\text{st}} \text{ AB and } 2^{\text{nd}} \text{ not AB and } 3^{\text{rd}} \text{ not AB}) \\
 &= 0.1 \times 0.9 \times 0.9 \\
 &= 0.081
 \end{aligned}$$

Blood groups of 3 people are independent of one another.

$$\begin{aligned}
 \text{(ii)} \quad & P(\text{exactly 1 AB}) \\
 &= 0.1 \times 0.9 \times 0.9 \times \frac{3!}{2!} \\
 &= 0.243
 \end{aligned}$$

Blood groups 'AB, not AB, not AB' can be arranged in $\frac{3!}{2!}$ ways.

$$\begin{aligned}
 \text{(iii)} \quad & P(\text{at least 1 AB}) \\
 &= 1 - P(\text{none AB}) \\
 &= 1 - P(1^{\text{st}} \text{ not AB and } 2^{\text{nd}} \text{ not AB and } 3^{\text{rd}} \text{ not AB}) \\
 &= 1 - (0.9 \times 0.9 \times 0.9) \\
 &= 0.271
 \end{aligned}$$

Using Method of Complementation.

$\square$

In parts (i) and (iii) of the following example, it is important to note that the events can occur in more than one order.

Example 26

Two cards are drawn at random from an ordinary deck of 52 cards without replacement. Find the probability that

- one card is a 'number card' and the other is a 'picture card',
- the two cards are of different suits,
- both cards are 'number cards' and the sum of the numbers shown on the two cards is 18.

Solution

$$(i) \quad P(\text{number card}) = \frac{40}{52}$$

$$P(\text{picture card}) = \frac{12}{52}$$

$$\text{Required probability} = \left(\frac{40}{52} \times \frac{12}{51} \right) + \left(\frac{12}{52} \times \frac{40}{51} \right) = \frac{80}{221}$$

OR

$$\text{Required probability} = \frac{40}{52} \times \frac{12}{51} \times 2! = \frac{80}{221}$$

The 2 cards can be drawn in either order: number card followed by picture card or vice versa.

$$(ii) \quad P(2 \text{ cards of different suits})$$

$$= \frac{52}{52} \times \frac{39}{51}$$

$$= \frac{13}{17}$$

After drawing 1st card (which can be any one of 52 cards), there are 39 cards of a suit different from that of the 1st card.

$$(iii) \quad P(\text{sum} = 18)$$

$$= P(\{(8, 10), (9, 9), (10, 8)\})$$

$$= \left(\frac{4}{52} \times \frac{4}{51} \right) + \left(\frac{4}{52} \times \frac{3}{51} \right) + \left(\frac{4}{52} \times \frac{4}{51} \right)$$

$$= \frac{11}{663}$$

□

Some probability problems can be solved in more than one way. In the following example, part (i) involving selection and arrangement can be solved using formulae from Chapter 1A Permutations and Combinations or this Chapter 1B Probability. Part (ii) involving “at least” (similarly “at most”) can be solved directly or by using method of complementation.

Example 27

Four people are to be chosen at random from five men and four women to form a team. One of the men is Max and one of the women is Wendy. Find the probability that the team consists of

- two men and two women,
- at least two women,
- Max or Wendy but not both,
- at least one man and at least one woman.

Solution

- (i) Method 1: (using Permutations and Combinations)

P(2 men and 2 women)

$$= \frac{\text{no. of ways to choose 2 out of 5 men and 2 out of 4 women}}{\text{no. of ways to choose 4 out of 9 people}}$$

$$= \frac{{}^5C_2 \times {}^4C_2}{{}^9C_4}$$

$$= \frac{10}{21}$$

Method 2: (using Probability)

$$P(2 \text{ men and } 2 \text{ women}) = \frac{5}{9} \times \frac{4}{8} \times \frac{4}{7} \times \frac{3}{6} \times \frac{4!}{2!2!} = \frac{10}{21}$$

When using probability method, $\frac{5}{9} \times \frac{4}{8} \times \frac{4}{7} \times \frac{3}{6}$ means the 4 people are chosen in that order, i.e. man, man, woman, woman. There is a need to consider other possible orders of selection, eg. man, woman, woman, man. Since the 2 men are classified as ‘men’, we cannot differentiate one ‘men’ from another, hence there is a need to divide by 2!. Similarly there is a need to divide by 2! for the 2 women.

- (ii) Method 1: (using Method of Complementation)

P(at least 2 women)

$$= 1 - P(\text{exactly 1 woman}) - P(\text{no women})$$

$$= 1 - \frac{{}^5C_3 \times {}^4C_1}{{}^9C_4} - \frac{{}^5C_4}{{}^9C_4}$$

$$= \frac{9}{14}$$

Method 2: (consider cases)

Case 1: Exactly 2 women

$$P(\text{exactly 2 women}) = \frac{10}{21}$$

Case 2: Exactly 3 women

$$P(\text{exactly 3 women}) = \frac{{}^5C_1 \times {}^4C_3}{{}^9C_4} = \frac{10}{63}$$

Case 3: Exactly 4 women

$$P(\text{exactly 4 women}) = \frac{{}^4C_4}{{}^9C_4} = \frac{1}{126}$$

$$\therefore P(\text{at least 2 women}) = \frac{10}{21} + \frac{10}{63} + \frac{1}{126} = \frac{9}{14}$$

- (iii) $P(\text{Max or Wendy in team but not both})$
 $= P(\text{Max in team but not Wendy}) + P(\text{Wendy in team but not Max})$
 $= \frac{{}^7C_3}{{}^9C_4} + \frac{{}^7C_3}{{}^9C_4}$
 $= \frac{5}{9}$
- (iv) $P(\text{at least 1 man and at least 1 woman})$
 $= 1 - P(\text{team consists all men}) - P(\text{team consists all women})$
 $= 1 - \frac{{}^5C_4}{{}^9C_4} - \frac{{}^4C_4}{{}^9C_4}$
 $= \frac{20}{21}$ □

Example 28

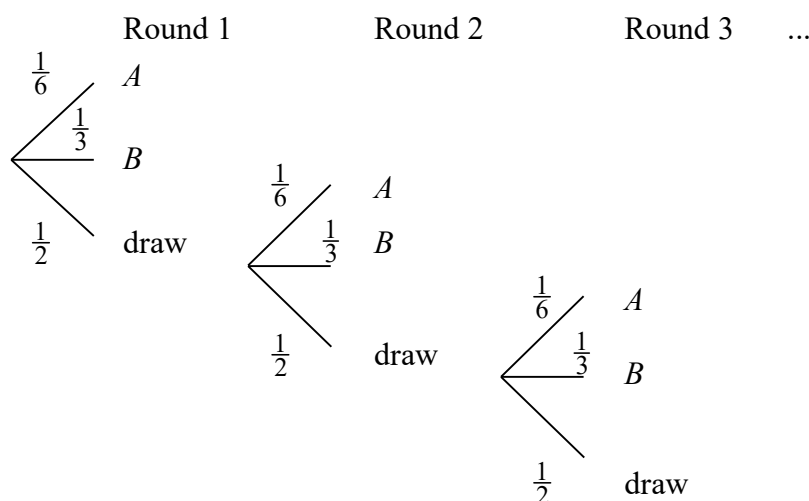
Ann and Bob take part in a tournament where in each round, $P(\text{Ann wins}) = \frac{1}{6}$,

$P(\text{Bob wins}) = \frac{1}{3}$ and $P(\text{draw}) = \frac{1}{2}$. The tournament continues until someone wins a round. Find the probability that

- Ann wins in the third round,
- Ann wins the tournament.
- Find the least value of n such that the probability of Ann winning in the n th round is less than 0.02.

Solution

- (i) Let A and B be events that Ann and Bob wins in a round respectively.



$P(\text{Ann wins in 3rd round})$

$$= \frac{1}{2} \times \frac{1}{2} \times \frac{1}{6}$$

$$= \frac{1}{24}$$

(ii) $P(\text{Ann wins tournament})$
 $= P(A) + P(\text{draw}, A) + P(\text{draw}, \text{draw}, A) + \dots$ ← Ann can win in 1st round, 2nd round, 3rd round, and so on.
 $= \frac{1}{6} + \left[\frac{1}{2} \times \frac{1}{6} \right] + \left[\left(\frac{1}{2} \right)^2 \times \frac{1}{6} \right] + \dots$
 $= \frac{\frac{1}{6}}{1 - \frac{1}{2}}$ ← Sum to infinity of a geometric series.
 $= \frac{1}{3}$

(iii) $P(\text{Ann wins in } n^{\text{th}} \text{ round})$
 $= P(\text{draw in } 1^{\text{st}} (n-1) \text{ rounds and Ann wins in } n^{\text{th}} \text{ round})$
 $= \left(\frac{1}{2} \right)^{n-1} \left(\frac{1}{6} \right)$
Hence $\left(\frac{1}{2} \right)^{n-1} \left(\frac{1}{6} \right) < 0.02$

Method 1: (using GC table)

n	$\left(\frac{1}{2} \right)^{n-1} \left(\frac{1}{6} \right)$
4	0.0208
5	0.0104 < 0.02
6	0.00521

Using GC, least $n = 5$

Method 2: (solve algebraically)

$$\left(\frac{1}{2} \right)^{n-1} < 0.12$$

$$(n-1) \ln \left(\frac{1}{2} \right) < \ln(0.12)$$

$$n-1 > \frac{\ln(0.12)}{\ln \left(\frac{1}{2} \right)}$$

$$n > 4.06$$

∴ least $n = 5$ □

Example 29

In a certain country, 1% of the population is infected with Omicron. A test kit is used in a screening exercise to identify Omicron victims. For someone who is infected with Omicron, there is a probability of 0.9 that the test kit will give a positive result. For someone who is not infected with the virus, there is a probability of 0.05 that the test kit will give a positive result.

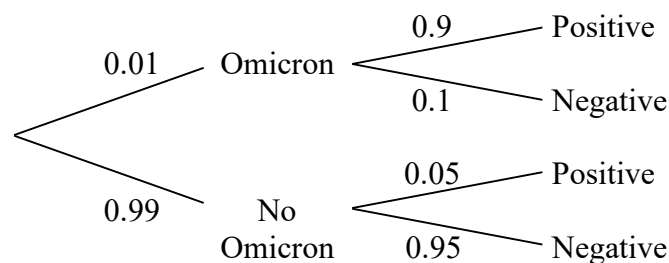
One person in that country is randomly selected and tested with the test kit.

- Find the probability that the person is not infected with Omicron and the test kit gives a positive result. [1]
- Find the probability that the test kit gives a positive result. [2]
- Given that the test kit gives a positive result, find the probability that the person is not an Omicron victim. Deduce the probability that the person is an Omicron victim given the test kit gives a positive result. [3]

Comment on the accuracy of the test in predicting the presence of the disease.
Suggest a possible reason for your answer.

Solution

(i)



$$\begin{aligned} P(\text{not infected and test positive}) &= 0.99 \times 0.05 \\ &= 0.0495 \end{aligned}$$

$$\begin{aligned} \text{(ii) } P(\text{test positive}) &= (0.01 \times 0.9) + 0.0495 \\ &= 0.0585 \end{aligned}$$

$$\begin{aligned} \text{(iii) } P(\text{not infected} \mid \text{test positive}) &= \frac{P(\text{not infected} \cap \text{test positive})}{P(\text{test positive})} \\ &= \frac{0.0495}{0.0585} \\ &= 0.8461538 \\ &= 0.846 \quad (3 \text{ s.f.}) \end{aligned}$$

$$\begin{aligned} \text{Hence } P(\text{infected} \mid \text{test positive}) &= 1 - 0.8461538 \\ &= 0.154 \quad (3 \text{ s.f.}) \end{aligned}$$

□

The test is not very accurate in predicting the presence of the disease. This is likely due to the low incidence of Omicron in the general population.

Summary

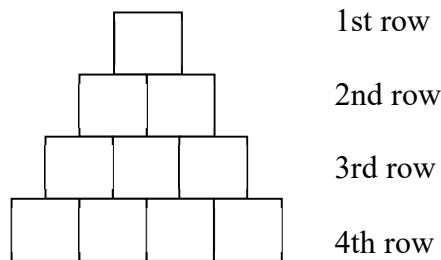
- If the sample space S consists of a finite number of equally likely outcomes, then $P(A) = \frac{n(A)}{n(S)}$, where $n(A)$ denotes the number of outcomes in the event A .
- For complementary event, $P(A') = 1 - P(A)$.
- Common useful results from Venn diagram:
 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
 $P(A) = P(A \cap B) + P(A \cap B')$
 $P(A' \cap B') = 1 - P(A \cup B)$
- Conditional Probability: $P(A | B) = \frac{P(A \cap B)}{P(B)}$ where $P(B) \neq 0$.
- Two events A and B are mutually exclusive if $A \cap B = \emptyset$, i.e., $P(A \cap B) = 0$.
- To show that two events A and B are independent, show either
 $P(A | B) = P(A)$ or
 $P(B | A) = P(B)$ or
 $P(A)P(B) = P(A \cap B)$.

 Statistics Tutorial 1

1. [MI20/Prelim P2/Q6]
Find the number of ways in which all the letters of the word FANTASTIC can be arranged if
 - (i) there are no restrictions, [2]
 - (ii) the first and last letters are consonants and the vowels are separated. [3]

2. [RVHS24/Prelim P2/Q6]
The number 396900 can be expressed as $2^2 \times 3^4 \times 5^2 \times 7^2$.
A factor of 396900 can be expressed in the form $2^a \times 3^b \times 5^c \times 7^d$ for non-negative integers a, b, c and d . For example, 3150 and 140 are factors of 396900 and they can be expressed as $2^1 \times 3^2 \times 5^2 \times 7^1$ and $2^2 \times 3^0 \times 5 \times 7$ respectively.
 - (a) State all the possible values for a . [1]
 - (b) Find the number of factors of 396900. [1]
 A factor of 396900 is chosen randomly.
 - (c) Find the probability that the chosen factor is divisible by 3 given that it is even. [3]

3. [DHS24/Prelim P2/Q7]
A bakery packs a box of 10 cupcakes comprising 2 red, 3 blue and 5 green ones. The cupcakes are indistinguishable except for their colour.
 - (a) Find the number of ways they can be given to 10 children, if each child is given exactly one cupcake. [1]
 - (b) The bakery packs the 10 cupcakes in a triangular arrangement as shown below.



Find the number of ways to arrange all the cupcakes such that all blue cupcakes are next to each other in the same row. [2]

For another box containing 10 brown identical cupcakes, the bakery decides to spell out the word DELECTABLE by using whipped cream to write one letter on each cupcake.

- (c) Find the number of ways the cupcakes can be arranged in a row in which the letters are **not** in alphabetical order. For example : the three letters 'BGZ' is arranged in alphabetical order while 'GBZ' is not. [2]
- (d) How many three-letter code word can be formed from the word DELECTABLE? [3]

Statistics Chapter 1: Probability

4. [CJC20/Prelim P2/Q9 (modified)]

Legox bricks come in different colours and sizes. Bricks of the same size and colour are identical. Jamie has 9 Legox bricks. The following table shows the various colours and sizes of Legox bricks that Jamie has.

	Small	Medium	Large
Green	0	0	2
Blue	1	1	1
Red	2	2	0

Jamie decides to use only 4 bricks to arrange in a row. Find the number of different possible arrangements if all 3 colours are used. [4]

5. [VJC20/Prelim P2/Q6]

A group of 8 people consists of 4 boys and 4 girls.

- (a) They go to a theatre and sit in the last row of 10 adjacent seats at random.
- (i) Find the number of ways in which they can be seated if they are not all seated together. [2]
- (ii) Given that they are not all seated together, find the probability that the boys are seated together and the girls are also seated together. [2]
- (b) At a dinner, the group sits around a square table which has two seats on each side.
- (i) Find the number of ways in which they can be seated.
- (ii) Find the number of ways in which they can be seated if there must be a boy and a girl on each side of the table. [3]
- (iii) There are 3 different desserts on the menu and every one selects a dessert each. Find the number of different selections that can be made by the group. [1]

6. [YIJC20/Prelim P2/Q5]

An interest group of 8 members consists of one married couple together with 4 other men and 2 other women. Find the number of ways that the 8 people can sit at a round table if

- (i) the married couple is to be seated together, [2]
- (ii) all women are to be separated, and the seats are numbered. [2]

An executive committee consisting of a Chairman, Vice-Chairman and Secretary is to be selected from the group of 8 members.

- (iii) Find the number of ways that the executive committee can be formed if at most one person from the married couple can be included. [3]

7. [DHS24/Prelim P2/Q6]

For events A and B , it is given that $P(A) = \frac{2}{5}$, $P(A|B') = \frac{3}{5}$ and $P(B|A') = \frac{5}{6}$. Find

- (a) $P(A' \cap B)$ [1]
- (b) $P(A' \cap B')$, [2]
- (c) $P(A \cap B)$. [3]

Determine if events A and B are independent. [1]

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8. [HCI20/Prelim P2/Q6]
- (a) Given that events A and B are independent, show that A' and B are also independent. [3]
- (b) A , B and C are events such that events A and B are independent and events A and C are mutually exclusive. It is given that $P(A)=0.3$, $P(B)=b$, $P(B \cap C)=0.15$ and $P(A' \cap B' \cap C')=0.2$. With the help of a Venn diagram, find the range of values for $P(A \cap B)$. [3]
9. [EJC24/Prelim P2/Q5 (modified)]
- A company sells consumer hobby drones, and they buy the rotor motor parts from two suppliers, AE Rotors and Brushless Inc. 25% of the motors are from AE Rotors and the rest are from Brushless Inc. The probability that a randomly chosen motor supplied by AE Rotors is faulty is $\frac{1}{20}$. The probability that a randomly chosen motor supplied by Brushless Inc is faulty is q .
- (a) Draw a tree diagram to illustrate the possibilities for a randomly chosen motor. You should use the following notations for the events:
- A for the motor is supplied by AE Rotors. B for the motor is supplied by Brushless Inc.
- F for the motor is faulty, and F' for its complement, i.e. the motor is not faulty. [1]
- It is known that the probability that a randomly picked motor that is faulty is supplied by Brushless Inc is $\frac{3}{8}$.
- (b) Use the above information to set up an equation involving q , and hence find the value of q . [2]
- (c) Three motors are randomly picked. Find the probability that all of them are supplied by AE Rotors and exactly one of them is faulty. [2]
10. [MI20/Prelim P2/Q8]
- A group of n people each takes turn to pick one card, with replacement, from a pack of cards.
- (a) Assume that there are 12 unique cards in a pack. The probability that all the people in the group picked different cards is denoted by P .
- (i) For the case where $n=3$, show that $P=\frac{55}{72}$. [2]
- (ii) Find P for the case where $n=4$ and deduce the smallest value of n such that $P < \frac{1}{2}$. [2]
- (iii) State, with a reason, the least possible value of n such that $P=0$. [2]

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- (b) Assume now that there are 120 unique cards in the pack. It is given that, for the case where $n = 12$, the probability that all the people in the group picked different cards is 0.56640, correct to 5 decimal places. Find the smallest value of n such that the probability that at least two people picked the same card exceeds $\frac{1}{2}$. [4]

11. [TJC20/Prelim P2/Q6]

Javier and Kelvin played a game with a pack of ten cards numbered 1, 2, 3, ..., 10. They take turns at drawing two cards each from the pack at random, one card at a time without replacement, and the two cards drawn are replaced before the next player draws. The player who first gets two cards whose sum is 10 wins the game. If he does not win, he replaces the two cards and the other player then takes his turn. The game carries on until a winner is decided.

On their first game, Javier gets to draw first.

- (i) Show that the probability that Javier wins the game on his first draw is $\frac{4}{45}$. [2]
- (ii) Find the probability that Kelvin wins the game. [3]

On their second game, they toss a biased coin to decide on the player to draw first. Using this biased coin, Kelvin is twice as likely as Javier to draw first.

- (iii) Find the probability that Javier wins the game. [3]

12. [DHS/2016/II/06]

John plays for his school's soccer team. There is a probability of 0.15 that he scores in a game and a probability of 0.3 that his parents are present at a game. When he scores in a game, there is a probability of 0.2 that his parents are present.

- (i) Show that the probability that he scores in a game when his parents are present is 0.1 [2]
- (ii) State, with justification, whether his parents' presence at a game will affect his chances of scoring in the game. [1]

Statistics Chapter 1: Probability

Answers

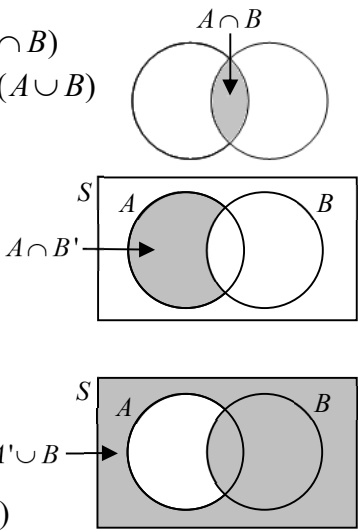
- 1(i) 90720 (ii) 10800
- 2(a) 0, 1, 2 (b) 135 (c) $\frac{4}{5}$
- 3(a) 2520 (b) 63 (c) 302399 (d) 247
4. 360
- 5(ai) 1693440 (aii) $\frac{1}{490}$ ~~(bi) 2304~~ ~~(bii) 6561~~
~~(bii)~~ ~~(biii)~~
- 6(i) 1440 (ii) 11520 (iii) 300
- 7(a) 0.5 (b) 0.1 (c) ~~not independent~~
~~0.25~~
- 8(b) $\frac{9}{140} \leq P(A \cap B) \leq \frac{3}{14}$
- 9(b) $q = 0.01$ (c) 0.00212
- 10(aii) smallest $n = 5$ (aiii) Least $n = 13$ (b) smallest $n = 14$
- 11(i) $\frac{4}{45}$ (ii) $\frac{41}{86}$ (iii) $\frac{127}{258}$
- 12(i) 0.1 (iii) Least $P(B' \cap C) = 0.35$, Greatest $P(B' \cap C) = 0.38$

Skill Sets

Statistics Chapter 1: Probability

s/n	Skills	Examples of questions involving the skills										
1	Permutation/Arrangement of • distinct • not all distinct objects in a row	[Lecture Notes Example 13(i)] Three boys and two girls in a class committee are to be arranged in a row in a photo-shoot. What is the number of ways to arrange them with no restriction? <u>Solution</u> No. of ways = $5! = 120$ [Lecture Notes Example 7] In how many ways can the letters of the word ‘STATISTICS’ be arranged? <u>Solution</u> <table><tr><td>S</td><td>S</td><td>S</td><td>T</td><td>T</td><td>T</td><td>I</td><td>I</td><td>A</td><td>C</td></tr></table> No. of ways = $\frac{10!}{3!3!2!} = 50400$	S	S	S	T	T	T	I	I	A	C
S	S	S	T	T	T	I	I	A	C			
2	Permutation/Arrangement of distinct objects in a circle	[Lecture Notes Example 11(a)] Two boys and four girls are to be seated at a round table. In how many ways can this be done if there are no restrictions? <u>Solution</u> No. of ways = $\frac{6!}{6} = 120$										
3	Combination/Selection of distinct objects without replacement	[Lecture Notes Example 9(a)(i)] There are 13 female and 7 male students in a class. A 4-member committee is to be selected. In how many ways can the committee be selected if there is no restriction to the selection? <u>Solution</u> No. of ways = ${}^{20}C_4 = 4845$										
4	Use Method of Grouping for objects that need “to be together”	[Lecture Notes Example 11(b)] Two boys and four girls are to be seated at a round table. In how many ways can this be done if the two boys must sit next to each other? <u>Solution</u> Group the two boys as 1 unit. Total no. of units = 5 No. of ways = $\frac{5!}{5} \times 2! = 48$										

s/n	Skills	Examples of questions involving the skills												
5	Use Method of Slotting for objects that need to be “separated”	[Lecture Notes Example 14(iii)] In how many ways can the letters of the word ‘INSTITUTION’ be arranged such that no two vowels are next to each other? <u>Solution</u> No. of ways = ${}^7C_5 \times \frac{6!}{2!3!} \times \frac{5!}{3!} = 25200$ Choose 5 out of 7 slots to place 5 vowels Arrange 5 vowels (with 3 ‘I’s) among themselves Arrange 6 consonants (with 2 ‘N’s and 3 ‘T’s) among themselves												
6	Use Method of Complementation when it is easier to consider the complement set/scenario	[Lecture Notes Example 13(iii)] Three boys and two girls in a class committee are to be arranged in a row in a photo-shoot. (i) What is the number of ways to arrange them with no restriction? (ii) What is the number of ways to arrange them so that the two girls are next to each other? (iii) What is the number of ways to arrange them so that the two girls are not next to each other? <u>Solution</u> (iii) No. of ways = (i) – (ii) = 120 – 48 = 72												
7	Consider cases Note: Questions requiring “at least” and “at most” usually requires consideration of cases	[Lecture Notes Example 9(a)(ii)] There are 13 female and 7 male students in a class. A 4-member committee is to be selected. In how many ways can the committee be selected if there must be at least two female students in the committee? <u>Solution</u> <table border="1" style="margin-left: auto; margin-right: auto;"> <thead> <tr> <th></th><th>Committee</th><th>No. of ways</th></tr> </thead> <tbody> <tr> <td>Case 1</td><td>2 females, 2 males</td><td>${}^{13}C_2 \times {}^7C_2 = 1638$</td></tr> <tr> <td>Case 2</td><td>3 females, 1 male</td><td>${}^{13}C_3 \times {}^7C_1 = 2002$</td></tr> <tr> <td>Case 3</td><td>All 4 females</td><td>${}^{13}C_4 = 715$</td></tr> </tbody> </table> No. of ways = $({}^{13}C_2 \times {}^7C_2) + ({}^{13}C_3 \times {}^7C_1) + {}^{13}C_4 = 4355$ [Tutorial Q1(ii)] Find the number of ways in which all the letters of the word FANTASTIC can be arranged if (ii) the first and last letters are consonants and the vowels are separated. <u>Solution</u> Case 1: Begin and end with ‘T’ No. of ways = $4! \times {}^5C_3 \times \frac{3!}{2!} = 720$ Case 2: Begin with ‘T’ but not end with ‘T’ (vice versa)		Committee	No. of ways	Case 1	2 females, 2 males	${}^{13}C_2 \times {}^7C_2 = 1638$	Case 2	3 females, 1 male	${}^{13}C_3 \times {}^7C_1 = 2002$	Case 3	All 4 females	${}^{13}C_4 = 715$
	Committee	No. of ways												
Case 1	2 females, 2 males	${}^{13}C_2 \times {}^7C_2 = 1638$												
Case 2	3 females, 1 male	${}^{13}C_3 \times {}^7C_1 = 2002$												
Case 3	All 4 females	${}^{13}C_4 = 715$												

s/n	Skills	Examples of questions involving the skills
		No. of ways = ${}^4C_1 \times 4! \times {}^5C_3 \times \frac{3!}{2!} \times 2! = 5760$ Case 3: Begin and end with non 'T' No. of ways = ${}^4C_2 \times 2! \times \frac{4!}{2!} \times {}^5C_3 \times \frac{3!}{2!} = 4320$ Total number of ways = 10800
8	Use combination/selection to find no. of ways to group objects, including groups of equal size	[Lecture Notes Example 9(b)] There are 13 female and 7 male students in a class. In how many ways can the 20 students be divided into two groups of (i) 8 and 12 students each? (ii) 10 students each? <u>Solution</u> (i) No. of ways = ${}^{20}C_8 \times {}^{12}C_{12} = 125970$ (ii) No. of ways = $\frac{{}^{20}C_{10}}{2!} = 92378$
9	Use Venn Diagram and apply the following as necessary to find probabilities [including finding greatest and least probabilities (see Tutorial 6(iii))]:	(Lecture Notes Example 20) The events A and B are such that $P(A) = 0.3$, $P(B) = 0.4$ and $P(A \cup B) = 0.5$. Find (i) $P(A \cap B)$, (ii) $P(A \cap B')$ (iii) $P(A' \cup B)$. <u>Solution</u> (i) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $\therefore P(A \cap B) = P(A) + P(B) - P(A \cup B)$ $= 0.3 + 0.4 - 0.5$ $= 0.2$ (ii) $P(A \cap B') = P(A \cup B) - P(B)$ $= 0.5 - 0.4$ $= 0.1$ (iii) Using (ii), $P(A' \cup B) = 1 - P(A \cap B')$ $= 1 - 0.1$ $= 0.9$ Alternatively, $P(A' \cup B) = P(A') + P(A \cap B)$ $= 0.7 + 0.2$ $= 0.9$ 
10	Identify/Prove events A and B as/are	(Lecture Notes Example 22) Draw a card at random from an ordinary deck of 52 cards. Let A be the event of drawing an ace. Let B be the event of drawing a spade. Determine if A and B are (i) mutually exclusive, (ii) independent. <u>Solution</u> (i) $P(A \cap B) = \frac{1}{52} \neq 0$ $\therefore A$ and B are not mutually exclusive. (ii) $P(A) = \frac{4}{52} = \frac{1}{13}$; $P(B) = \frac{13}{52} = \frac{1}{4}$

s/n	Skills	Examples of questions involving the skills
	$P(B A) = P(B)$	Since $P(A \cap B) = \frac{1}{52} = \frac{1}{4} \times \frac{1}{13} = P(A)P(B)$, $\therefore A$ and B are independent events.
11	Identify, and evaluate conditional probabilities using $P(A B) = \frac{P(A \cap B)}{P(B)}$	[Lecture Notes Example 21(ii)] Two ordinary dice are thrown. Find the probability that the numbers on the two dice are equal, given that the sum of the numbers on the two dice is 8. <u>Solution</u> Let A be event that sum of two dice is 8. Let B be event that numbers on two dice are equal. $P(B A) = \frac{P(A \cap B)}{P(A)} = \frac{P(\{(4,4)\})}{P(\{(2,6),(3,5),(4,4),(5,3),(6,2)\})} = \frac{\frac{1}{36}}{\frac{5}{36}} = \frac{1}{5}$
12	Use tree diagrams to find probabilities	[Lecture Notes Example 24(i), (ii)] A bag contains two red balls and three green balls. Three balls are drawn from the bag, one after another without replacement. Find the probability that (i) exactly one of the balls drawn is red, (ii) exactly one red ball is drawn, given that the first ball drawn is green. <u>Solution</u> Let R be 'a red ball drawn' Let G be 'a green ball drawn'. From tree diagram, (i) $P(\text{exactly 1 red})$ $= P(R_1 \cap G_2 \cap G_3) + P(G_1 \cap R_2 \cap G_3) + P(G_1 \cap G_2 \cap R_3)$ $= \left(\frac{2}{5} \times \frac{3}{4} \times \frac{2}{3}\right) + \left(\frac{3}{5} \times \frac{2}{4} \times \frac{2}{3}\right) + \left(\frac{3}{5} \times \frac{2}{4} \times \frac{2}{3}\right)$ $= \frac{3}{5}$ (ii) $P(\text{exactly 1 red} 1^{\text{st}} \text{ ball green})$ $= \left(\frac{2}{4} \times \frac{2}{3}\right) + \left(\frac{2}{4} \times \frac{2}{3}\right)$ $= \frac{2}{3}$

s/n	Skills	Examples of questions involving the skills																																																											
13	Construct a table of possible outcomes to calculate probabilities	[Lecture Notes Example 18(iv)] Two ordinary fair dice are thrown. Find the probability that the sum of the numbers on the two dice exceeds 8. <u>Solution</u> <table><tr><th colspan="2">Sum</th><th colspan="6">Number on the first die</th></tr><tr><th colspan="2"></th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th></tr><tr><th rowspan="6">Number on the second die</th><th>1</th><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><th>2</th><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr><tr><th>3</th><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr><tr><th>4</th><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td></tr><tr><th>5</th><td>6</td><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td></tr><tr><th>6</th><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td><td>12</td></tr></table> $P(\text{sum of two dice exceeds } 8) = \frac{10}{36} = \frac{5}{18}$	Sum		Number on the first die								1	2	3	4	5	6	Number on the second die	1	2	3	4	5	6	7	2	3	4	5	6	7	8	3	4	5	6	7	8	9	4	5	6	7	8	9	10	5	6	7	8	9	10	11	6	7	8	9	10	11	12
Sum		Number on the first die																																																											
		1	2	3	4	5	6																																																						
Number on the second die	1	2	3	4	5	6	7																																																						
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	3	4	5	6	7	8	9																																																						
	4	5	6	7	8	9	10																																																						
	5	6	7	8	9	10	11																																																						
	6	7	8	9	10	11	12																																																						
14	Use permutations and combinations to calculate probabilities (including conditional probabilities)	[Lecture Notes Example 27(i)] Four people are to be chosen at random from five men and four women to form a team. One of the men is Max and one of the women is Wendy. Find the probability that the team consists of two men and two women. <u>Solution</u> $P(2 \text{ men and } 2 \text{ women})$ $= \frac{\text{No. of ways to choose 2 out of 5 men and 2 out of 4 women}}{\text{No. of ways to choose 4 out of 9 people}}$ $= \frac{{}^5C_2 \times {}^4C_2}{{}^9C_4}$ $= \frac{10}{21}$ N-19 N-38																																																											
15	Use the Learning Objectives on page 1 and the Summaries on pages 18 and 36 of the Lecture Notes to review the key concepts and learning points of the chapter.																																																												

Supplementary Exercise

1. IJC/2018/II/Q6

Find the number of ways in which the letters of the word APPRECIATE can be arranged if

- (i) vowels (A, E, I) and consonants (P, R, C, T) must alternate, [2]
- (ii) between the two Es, there are exactly two other letters and at least one of which must be an A. [3]

2. AJC/2017/II/Q9

Four families arrive at Science Centre together. Mr and Mrs *A* brought their 2 children while Mr *B* brought his 2 children. Mr and Mrs *C* brought their 3 children while Mrs *D* brought her only child. All these 14 people have to go through a gate one at a time to enter the centre.

- (i) In how many different ways can they go through the gate if each family goes in one after another? [2]

There are two experiments at the *Science Magic Experience* station.

- (ii) In one experiment, participants are to be in groups of twos or threes. In how many different ways can the 8 children from the four families be grouped among themselves? [3]
- (iii) In another experiment, the four families have to hold hands to form two separate circles of equal size to experience a science phenomenon. Each circle must have exactly four children and members of the same family must be in the same circle. Find the number of ways of arranging these 14 people in the two circles such that there is no more than one adult between any two children. [3]

3. ACJC/2015/II/Q7

- (a) A teacher wants to set a class test for her class of 25 students. She plans to give each student the same questions, but have each student's questions appear in a different order. Find the least number of questions she must set. [2]
- (b) Sally has 12 bars of chocolates in four different flavours as summarised in the table below.

flavour	white	milk	dark	hazelnut
number	4	3	4	1

Sally intends to give one bar of chocolate each to three of her best friends in school.

- (i) Find the number of ways she can do it. [3]

- (ii) Sally remembers that one particular friend amongst the three has a nut allergy, and should not be given the hazelnut chocolate bar. Find the number of ways she can now give her friends the chocolates. [2]

4. **NYJC/2016/II/07**

A team of 5 men and 5 women is to be picked from 8 men and 9 women such that two of the 9 women, Ann and Lucy, must both be selected or not at all. Find the number of ways in which this can be done. [2]

Assume now the team is selected and Ann, Carrie and Lucy are included.

- (i) The selected team is to form a queue. Find the number of possible arrangements if Ann and Lucy are to occupy both the second and the sixth positions and no two people of the same gender are to stand next to each other. [3]
- (ii) On another occasion, the selected team is required to be seated at a round table with 10 chairs of different colours. If only Carrie can be seated between Ann and Lucy, find the number of possible arrangements. [3]

5. **HCI/2017/II/Q7**

Mandy has 10 beads, of which 5 are spherical and 5 are cubical, each of different colours. She wishes to decorate a card by forming a circle using 8 of the 10 beads. Find the number of ways Mandy can arrange the beads if

- (i) there are no restrictions, [1]
- (ii) 3 particular beads are used and not all are next to one another, [3]
- (iii) spherical beads and cubic beads must alternate. [3]

6. **NJC/2017/II/Q5**

A delegation of four students is to be selected from five badminton players, m floorball players, where $m > 3$, and six swimmers to attend the opening ceremony of the 2017 National Games. A pair of twins is among the floorball players. The delegation is to consist of at least one player from each sport.

- (i) Show that the number of ways to select the delegation in which neither of the twins is selected is $k(m - 2)(m + 6)$, where k is an integer to be determined. [3]
- (ii) Given that the number of ways to select a delegation in which neither of the twins is selected is more than twice the number of ways to select a delegation which includes exactly one of the twins, find the least value of m . [2]

The pair of twins, one badminton player, one swimmer and two teachers, have been selected to attend a welcome lunch at the opening ceremony. Find the number of ways in which the group can be seated at a round table with distinguishable seats if the pair of twins is to be seated together and the teachers are separated. [3]

7. **MI/2017/II/Q10**

- (a) Find the number of ways in which the letters of the word MILLENNIUM can be arranged if
- (i) there are no restrictions, [1]
 - (ii) the first and last letters are the same, and the letters E and U must be separated. [2]

Four letters are randomly selected from the letters of the word MILLENNIUM to form a code word. Find the number of possible code words that can be formed. [2]

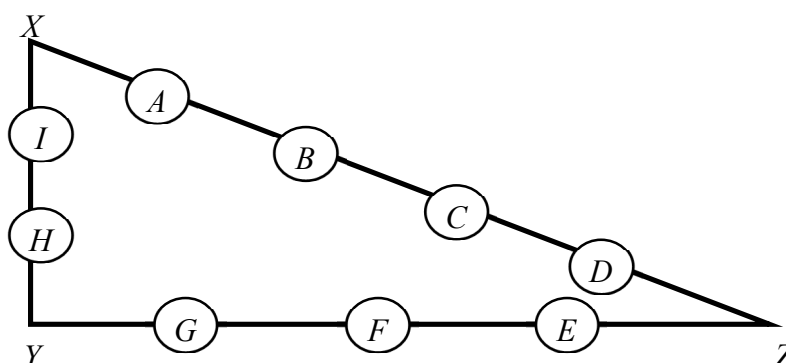
- (b) Mr See (together with his wife and daughter) and Mrs Saw (together with her husband and two sons) came to visit their former teacher Mdm Rain during Millennium Institute's Homecoming Day. Find the number of ways Mr See and his family, Mrs Saw and her family, and Mdm Rain can be arranged if
- (i) they are around a table with ten indistinguishable chairs, such that the children are seated together. [2]
 - (ii) the two empty chairs are removed and Mr See's daughter is seated between her parents and the See family are to be seated directly opposite Mdm Rain. [3]

8. **[EJC Prelim 9758/2022/02/Q8]**

In an art installation, three points on the floor, X , Y and Z , were connected using a rope to form a triangle XYZ . 9 bulbs, labelled $A, B, C, D, E, F, G, H, I$, are then to be placed along the rope to improve the aesthetic appearance.

- (a) The 9 bulbs are first distributed into 3 groups, namely Group XY , Group YZ and Group ZX . Find the number of ways the bulbs can be distributed if each group consists of at least 2 bulbs. [2]

In one particular distribution from part (a), the bulbs from Group XY , Group YZ and Group ZX are placed on the sides XY , YZ and ZX of the triangle XYZ respectively. The bulbs are arranged as shown in the diagram below.



- (b) By connecting the bulbs in this arrangement using additional rope, we can form a smaller triangle within the triangle XYZ . For example, the bulbs B, I, G form a smaller triangle when connected. How many different smaller triangles can be formed? [2]
- (c) Each bulb can be programmed to produce either red, blue or green light when it is switched on. Find the number of ways the colours can be programmed so that all the 3 colours are used. [3]

In another art installation, 9 bulbs, labelled $A, A, B, B, C, C, D, E, F$, are laid on the floor in a circular arrangement.

- (d) Find the number of possible arrangements if the 2 A 's are separated and the 2 B 's are separated. [3]

9. RI/2017/II/Q10

- (a) It is given that the probability that 21 randomly chosen people were all born on different days of the year is 0.55631, correct to 5 decimal places.

Find the probability that in a random sample of 22 people, there are at least 2 people with the same date of birth. [3]

[You may assume there are 365 days in a year and the probability that a person is born on any of the 365 days is the same.]

- (b) A soccer team consists of 1 goalkeeper, 4 defenders, 4 midfielders and 2 forwards.

Country N has a squad of 3 goalkeepers, 6 defenders, 9 midfielders and 4 forwards.

- (i) How many different soccer teams can be formed by country N? [2]

One of the defenders and one of the midfielders in the squad are twin brothers.

- (ii) How many different teams can be formed which include at most one of the twin brothers? [3]

The following table shows the dates of birth of the 22 players in the squad of country N:

Jersey Number	Position	Date of birth
1	Goalkeeper	29 October
2	Defender	3 May
3	Defender	17 July
4	Defender	15 May
5	Defender	14 December
6	Midfielder	12 October
7	Midfielder	15 May
8	Forward	10 May
9	Forward	1 July
10	Midfielder	1 April
11	Midfielder	29 October

Jersey Number	Position	Date of birth
12	Midfielder	15 August
13	Defender	11 July
14	Midfielder	29 March
15	Defender	22 October
16	Midfielder	13 March
17	Forward	29 November
18	Goalkeeper	20 December
19	Midfielder	5 February
20	Midfielder	1 March
21	Forward	27 March
22	Goalkeeper	31 October

- (iii) Find the probability that the team formed by country N contains no players with the same date of birth. [4]

10. SAJC/2016/II/07

- (a) For events X and Y , it is given that $P(X \cup Y) = \frac{5}{8}$, $P(X \cap Y') = \frac{7}{24}$ and

$$P(X'|Y) = \frac{9}{16}.$$

- (i) Find $P(X' \cap Y)$, [3]

- (ii) Find $P(X)$ and determine if the events X and Y' are independent. [3]

- (b) The Mathematics department consists of 5 female teachers and 6 male teachers. After a meeting, the department went to a nearby food court for lunch. Due to the lunch crowd, they only managed to find a circular table for 6 and a long table with a row of 5 seats as shown below. The seats at both tables are fixed and cannot be rearranged.

Ms Koh was among the Mathematics teachers who attended the lunch.



- (i) Find the probability that Ms Koh is seated between 2 male teachers. [4]
- (ii) Given that Ms Koh is seated between 2 male teachers, find the probability that the male and female teachers alternate at both tables. [3]

11. CJC/2016/II/05

A bag contains four red and eight blue balls of which two of the red balls and six of the blue balls have the number “0” printed on them. The remaining balls have the number “1” printed on them. Three balls are randomly drawn from the bag without replacement.

- (i) Show that the probability that at least one blue ball is drawn is $\frac{54}{55}$. [1]

Find the probability that

- (ii) at least one ball of each colour is drawn, [2]
- (iii) the sum of the numbers on the balls drawn is at least two. [3]

12. MJC/2016/II/09

- (a) For events A and B , it is given that $P(A) = \frac{1}{4}$ and $P(B) = \frac{1}{2}$.

- (i) Given that $P(A'|B) = \frac{3}{4}$, determine whether events A and B are independent and calculate $P(A \cup B)$. [3]

- (ii) For a third event C , it is given that $P(C|A) = \frac{2}{3}$.
Find the value of $P(A \cap C)$. [2]

- (b) Find the number of ways in which the word EVERYDAY can be arranged if
- (i) all the vowels (A, E) must be together and the two 'Y's must be separated, [3]
 - (ii) the repeated letters E and Y must appear symmetrical about the centre of the word (e.g. EVRYYYDAE, YVERDEAY). [2]

13. PJC/2017/II/Q10

For events A and B , it is given that $P(A) = \frac{11}{20}$ and $P(B) = \frac{1}{2}$.

- (i) Find the greatest and least possible values of $P(A \cap B)$. [2]

It is given in addition that $P(B | A') = \frac{7}{9}$.

- (ii) Find $P(A \cup B)$. [2]

- (iii) Determine if A and B are independent events. Justify your answer. [2]

- (iv) Given another event C such that $P(C) = \frac{2}{5}$, $P(A \cup B \cup C) = \frac{19}{20}$,

$$P(A \cap B \cap C) = \frac{1}{10} \text{ and } P(A \cap C) = 2P(B \cap C), \text{ find } P(A \cap C). \quad [3]$$

14. HCI/2016/II/07

A group of ten people consists of four single women, two single men and 2 couples. The ten people are arranged randomly in a circle.

- (i) Find the probability that the four single women are all separated. [2]
- (ii) Find the probability that either the four single women are next to one another or the two single men are next to each other but not both. [3]
- (iii) One of the ten people left the group and the remaining nine decided to sit at a round table with ten identical chairs equally spaced around the table. The chairs are decorated such that every alternate chair is tied with an identical chair sash. Given that the nine people have no preference to which seat to take, find the number of possible seating arrangements. [2]

15. VJC/2017/II/Q5

John and Peter play a game of chess. It is equally likely for either player to make the first move. If John makes the first move, the probability of him winning the game is 0.3 while the probability of Peter winning the game is 0.2. If Peter makes the first move, the probability of him winning the game is 0.5 while the probability of John winning the game is 0.4. If there is no winner, then the game ends in a draw.

- (i) Find the probability that Peter made the first move given that he won the game. [3]
- (ii) John and Peter played a total of three games. Assuming that the results of the three games are independent, find the probability that John wins exactly one game. [3]

16. EJC/2018/II/Q6

- (a) Two events A and B occur with probabilities a and b respectively.
Let $c = P(A \cap B)$.
- (i) Find the minimum value of c , terms of a and b , if $a + b > 1$. [2]
- (ii) Find the minimum value of c if $a = 0.6$ and $b = 0.2$. [1]
- (b) For independent events A and B , prove that the events A' and B' are also independent. [3]

17. DHS/2018/II/Q7

During a symposium, 4 boys and 8 girls are divided into 4 groups of three each for discussion. How many ways are there to divide the 12 participants such that each group consists of exactly 1 boy? [2]

After the discussion, all members of the 4 groups sit at random at a round table.

Find the probability that

- (i) the 3 members in each group are next to each other, [2]
- (ii) every boy is separated from each other by exactly 2 girls. [3]

For the 12 participants, events A and B are defined by

A : every boy is separated from each other by exactly 2 girls; and

B : none of the boys are seated next to each other.

- (iii) Determine if the events A and B are independent. [2]

18. VJC/2012/II/Q10

Two fair n -sided dice, each with faces numbered '1', '2', ... ' n ', (where $n \geq 4$) are tossed in an experiment. The number that appears on each die is defined as the number on the face that is in contact with the table. If the numbers that appear on the 2 dice are different, the score is taken as the larger of the two numbers that appears and the experiment ends. If the same number appears on both dice, the two dice are rolled again until two different numbers are obtained.

A trial is defined as a toss of two dice.

- (a) If $n = 6$,
- (i) show that the probability of obtaining a score of 3 with two trials is $\frac{1}{54}$, [3]
- (ii) find the probability that the score is 3, [3]
- (iii) find the probability that there are exactly four trials, given that the score is 3. [2]
- (b) Show that the probability that the score is '3' can be expressed as $\frac{a}{n(n-b)}$, where a, b are integers to be determined. [3]
- Hence, find the greatest number of faces on each die such that the probability of obtaining a score of 3 is at least 5%. [2]

19. EJC/2018/II/Q11

A bag contains 5 cards with the letter 'A', 3 cards with the letter 'B' and 2 cards with the letter 'C'.

A A A A A B B B C C

The 10 cards are arranged at random in a row to form a letter sequence.

For example, AABBCAAACB is a possible letter sequence.

- (i) Find the number of possible letter sequences. [1]
- (ii) Find the number of possible letter sequences if no two 'B's are next to each other **and** no two 'C's are next to each other. [4]
- (iii) Find the probability that the first two letters are identical given that the second letter is **not** an 'A'. [4]

The 10 cards are now arranged at random in a circle.

- (iv) Find the probability that no two 'A's are next to each other. [3]

20. NJC/2015/II/Q10

In a particular country, cultured milk are manufactured, in bottles, by two companies, Yacoat and Vitergent. 60% of all cultured milk in that country are manufactured by Yacoat while the remaining 40% are manufactured by Vitergent. Both companies manufacture the cultured milk in three different flavours, "Apple", "Grape" and "Orange", and the percentage of each flavor manufactured by each of the two companies is as follows,

	Apple	Grape	Orange
Yacoat	50%	30%	20%
Vitergent	100p%	10%	100q%

where p and q are real constants. A bottle of cultured milk is randomly selected.

- (a) Find the probability that
 - (i) it is grape-flavoured. [2]
 - (ii) it was manufactured by Yacoat, given that it is grape-flavoured. [1]
- (b) Suppose that the probability that it is apple-flavoured and is manufactured by Vitergent is 0.1 less than the probability that it is orange-flavoured and manufactured by Vitergent. By setting up two equations involving p and q , find the values of p and q . [3]

Assume now that $p = 0.4$ and $q = 0.5$, and that two bottles of cultured milk are randomly selected.

- (c) Events A and B are defined as follows:

A : One of them is manufactured by Yacoat and the other by Vitergent.

B : Both bottles are apple-flavoured.

Find $P(A \cup B)$. [4]

21. NJC/2018/II/Q6

A school's concert band comprises 24 woodwind players, n brass players and 10 percussion players. $\frac{1}{3}$ of all woodwind players, $\frac{2}{5}$ of all brass players and $\frac{4}{5}$ of all percussion players are Senior High students, while the rest are Junior High students. No student in the band plays more than one type of instrument.

One student from the concert band is selected at random. Find, in terms of n , the probability that the student is neither a percussion player nor a Senior High student. [3]

Suppose instead that two students from the concert band are randomly selected. It is given that the probability that one of them is a Senior High student and the other is a Junior High student is $\frac{1}{2}$. Show that

$$n^2 + pn + q = 0$$

for some integer constants p and q to be determined, and hence find the value of n . [3]

22. MJC/2018/II/Q8

A circular disc is divided into ten equal sectors where five sectors are coloured blue, three sectors are coloured green and two sectors are coloured red.

In a game, a player is given a maximum of three times to spin a pointer pivoted at the centre of the disc.

- If the pointer lands on a blue sector, the player wins \$2 and the game continues.
- If the pointer lands on a green sector, the game ends and the player loses all of his winnings.
- If the pointer lands on a red sector in the first spin, the game ends and the player wins the grand prize of \$20.
- If the pointer lands on a red sector in the second or third spin, the player wins double the amount of his total winnings from all his previous spins, and the game continues. For example, if the pointer lands on a blue sector in the first spin and on a red sector in the second spin, the total winnings for the first two spins will be \$6.

- (i) Construct a probability tree showing this information. [3]
- (ii) Find the exact probability that the player wins \$12 or more when the game ends. [2]
- (iii) Find the exact probability that the pointer in the second spin lands on a blue sector given that the player wins \$12 or more when the game ends. [2]

- (iv) Suppose there is no limit to the number of times a player can spin, find the exact probability that the player has no spins in which the pointer lands on a red sector and wins nothing when the game ends. [2]

23. DHS/2017/II/Q10

- (a) The word DISTRIBUTION has 12 letters.
- (i) Find the number of different arrangements of the 12 letters that can be made. [1]
- (ii) Find the number of different arrangements which can be made if there are exactly 8 letters between the two Ts. [3]

One of the Is is removed from the word and the remaining letters are arranged randomly.

- (iii) Find the probability that no adjacent letters are the same. [4]
- (b) The insurance company Adiva classifies 10% of their car policy holders as 'low risk', 60% as 'average risk' and 30% as 'high risk'. Its statistical database has shown that of those classified as 'low risk', 'average risk' and 'high risk', 1%, 15% and 25% are involved in at least one accident respectively.

Find the probability that

- (i) a randomly chosen policy holder is not involved in any accident if the holder is classified as 'average risk', [1]
- (ii) a randomly chosen policy holder is not involved in any accident, [2]
- (iii) a randomly chosen policy holder is classified as 'low risk' if the holder is involved in at least one accident. [2]

24. MI/2017/II/Q6

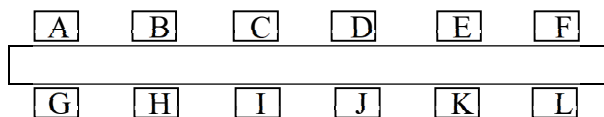
In another survey conducted by the National Eye Centre, it was found that $p\%$ are boys and the remaining are girls. The probability that a randomly chosen boy wears spectacles is 0.3 and the probability that a randomly chosen girl wears spectacles is 0.24.

- (i) Find the value of p , given that the probability that a randomly chosen child wears spectacles is 0.267. [2]
- (ii) For a general value of p , the probability that a randomly chosen child that wears spectacles is a girl is denoted by $f(p)$. Show that $f(p) = \frac{4(100-p)}{(400+p)}$. Prove by differentiation that f is a decreasing function for $0 \leq p \leq 100$, and explain what this statement means in the context of the question. [5]

25. SRJC/2016/II/05

Nicole decides to celebrate her birthday with 9 boys and 2 girls whose names are Vanessa and Sally.

- (a) (i) They have a dinner at a restaurant that can only offer them a rectangular table as shown in the following diagram, with seats labelled A to L as shown.



Find the number of ways in which at least one girl must be seated at the seats A, F, G and L.

- (ii) Find the number of ways in which they can sit if instead, the restaurant offers them 2 indistinguishable round tables of 6.
- (iii) After the dinner, they went for a movie together. They bought tickets for seats in a row. Find the number of ways where Nicole and Vanessa must be seated together but not Sally.
- (b) After the celebration, Nicole plays a card game with Vanessa. The pack of 20 cards are numbered 1 to 20. The two friends take turns to draw a card from the pack. If a prime number is drawn, the player wins the game. If a composite number (4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20) is drawn, the player loses the game and the other player wins. If the number '1' is drawn, the card is returned and the other player draws the next card. Nicole draws the first card. Find the probability of her winning the game.

Answers

- 1(i) 3600 (ii) 32760
 2(i) 829440 (ii) 385 (iii) 20736
 3(a) 5 (bi) 54 (bii) 45
 4. 3136 (i) 1440 (ii) 100800
 5(i) 226800 (ii) 90720 (iii) 3600
 6(i) $k = 15$ (ii) 9; 144
 7(ai) 226800 (ii) 15120; 876 (bi) 15120 (bii) 48
 8(a) 11508 (b) 79 (c) 18150 (d) 2880
 9(a) 0.476 (bi) 34020 (bii) 23940 (biii) $\frac{16}{27}$
 10(ai) $\frac{3}{16}$ (aii) Independent (bi) $\frac{3}{11}$ (bii) $\frac{1}{63}$
 11(i) $\frac{54}{55}$ (ii) $\frac{8}{11}$ (iii) 0.236
 12(ai) Independent; $\frac{5}{8}$ (aii) $\frac{1}{6}$ (bi) 720 (bii) 288
 13(i) $\frac{1}{20}$; $\frac{1}{2}$ (ii) $\frac{9}{10}$ (iii) Not independent (iv) 0.3
 14(i) $\frac{5}{42}$ (ii) $\frac{5}{21}$ (iii) 725760
 15(i) $\frac{5}{7}$ (ii) 0.444
 16(ai) $a + b - 1$ (ii) 0
 17. 2520 (i) 0.000195 (ii) $\frac{1}{165}$ (iii) Not independent
 18(aii) $\frac{2}{15}$ (iii) $\frac{5}{1296}$ (b) $a = 4, b = 1$, greatest $n = 9$
 19(i) 2520 (ii) 966 (iii) $\frac{8}{45}$ (iv) $\frac{1}{126}$
 20(ai) 0.22 (ii) $\frac{9}{11}$ (b) $p = 0.325, q = 0.575$ (c) $\frac{1489}{2500}$
 21. $\frac{3n+80}{5n+170}$; $p = -5, q = -750$; 30
 22(ii) $\frac{27}{100}$ (iii) $\frac{5}{27}$ (iv) $\frac{3}{5}$
 23(ai) 39916800 (ii) 1814400 (iii) 0.673 (bi) 0.85 (ii) $\frac{417}{500}$ (iii) $\frac{1}{166}$
 24(i) 45
 25(ai) 357073920 (ii) 6652800 (iii) 65318400 (b) $\frac{3}{7}$